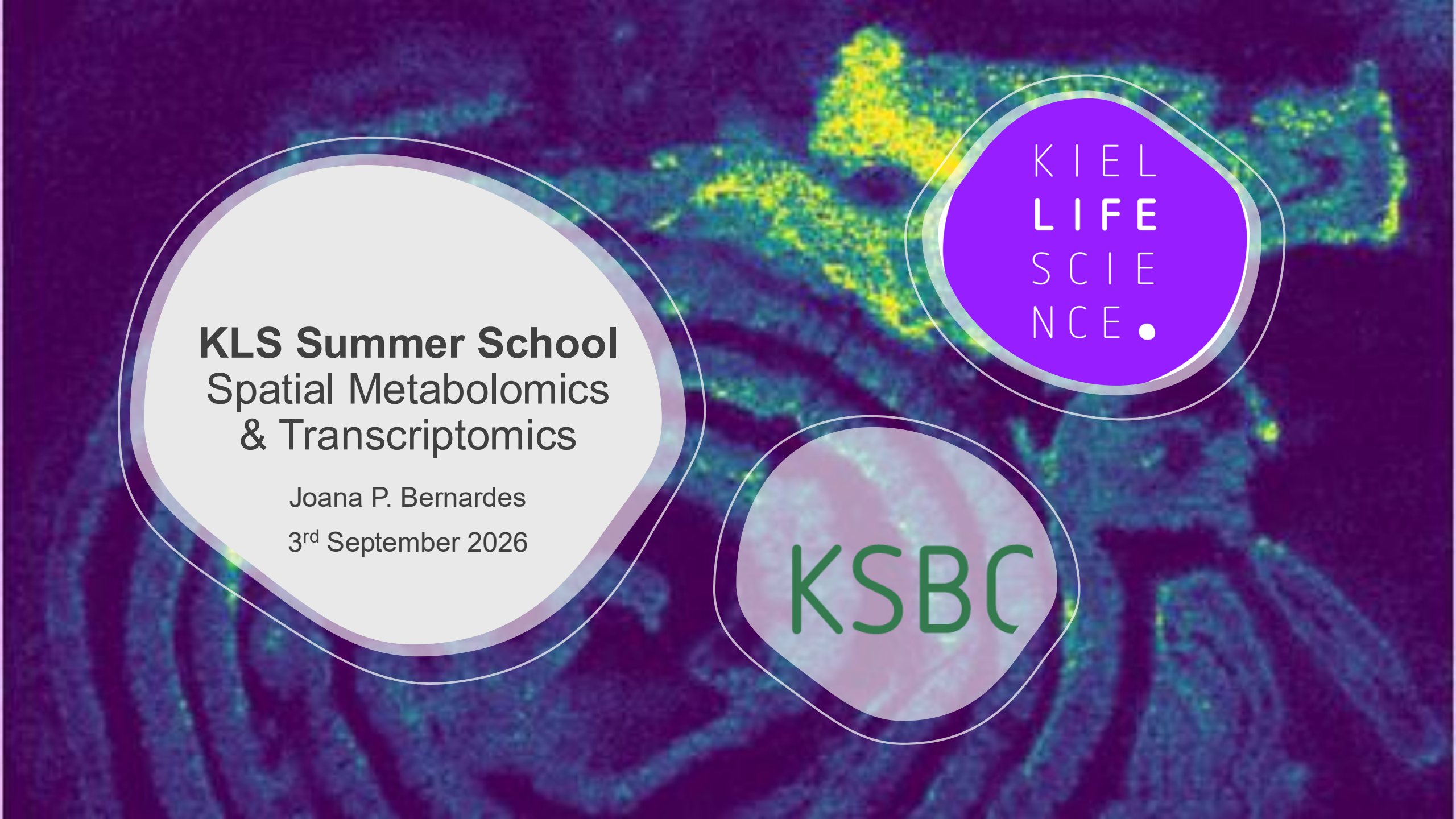




KLS Summer School Spatial Metabolomics & Transcriptomics

Joana P. Bernardes
3rd September 2026

K I E L
L I F E
S C I E
N C E •

KSBC

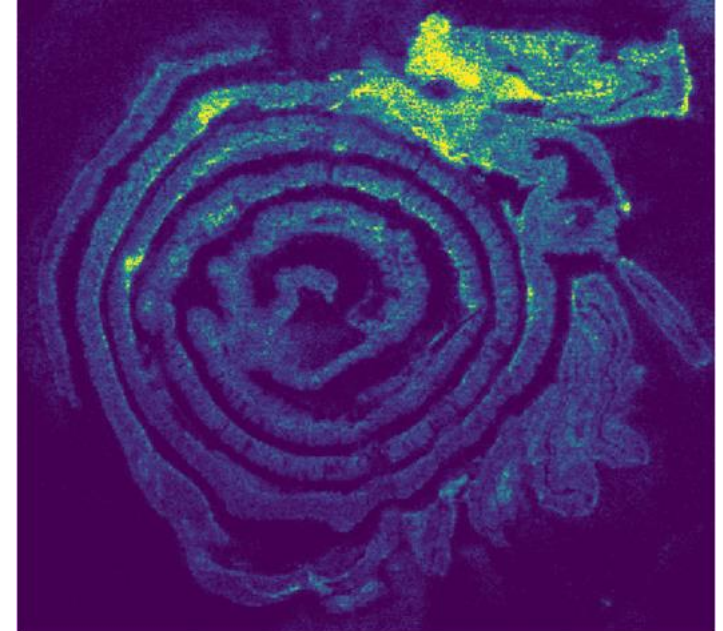
Agenda: 2nd September

13:00-13:15- General information about data analysis

13:15-15:30: Separate into groups

- 1- 3rd floor metabolomics center
- 2- EG for cryotome
- 3- Stay in 4th to set up R environment

```
conda env create -f r-Spatial_Meta_environment.yml
```



Agenda: 3rd September

09:00-09:30: Spatial Metabolomics (SM) & Transcriptomics (ST)

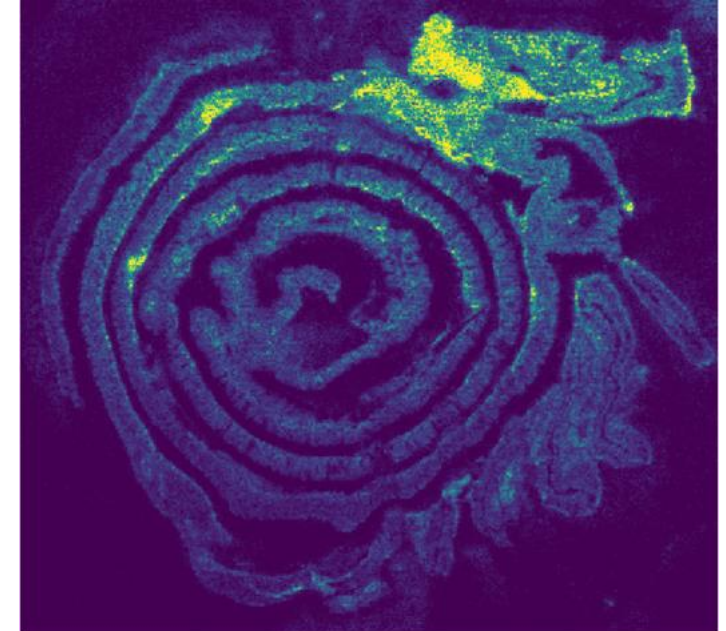
09:30-10:30

Spatial Metabolomics (MALDI)

Spatial Transcriptomics (Xenium)

10:30-10:45: Coffee break

10:45-12:15: Integration of Datasets



Dataset for today's class: Mouse swiss roll colon

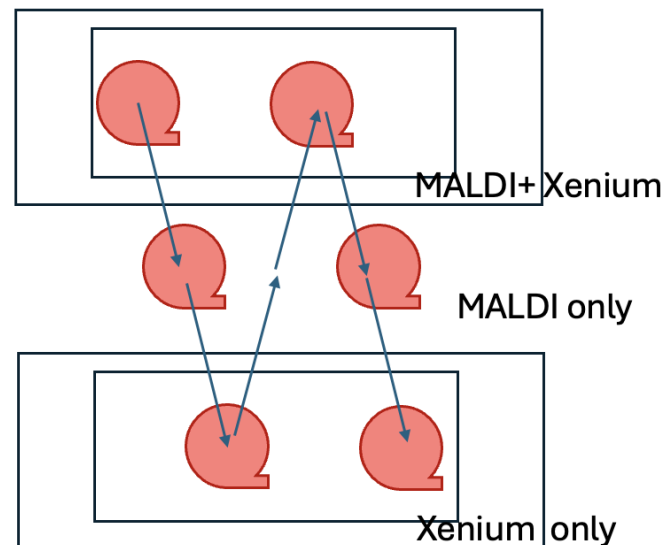
WT type mice swiss roll preserved in CMC.

In a Xenium slide we will lay 6 consecutive cuts

- MALDI** - 1 cut lipid panel
- 1 cut small molecule panel
- 1 cut nothing

Run Xenium custom mouse panel

Xenium V1 with custom panel for intestinal epithelium (Inflammation & Metabolism).



Packages for today's class

R-based class

SM: Cardinal

ST: Seurat

Integration: SpaMTP

Python:

MALDIpy

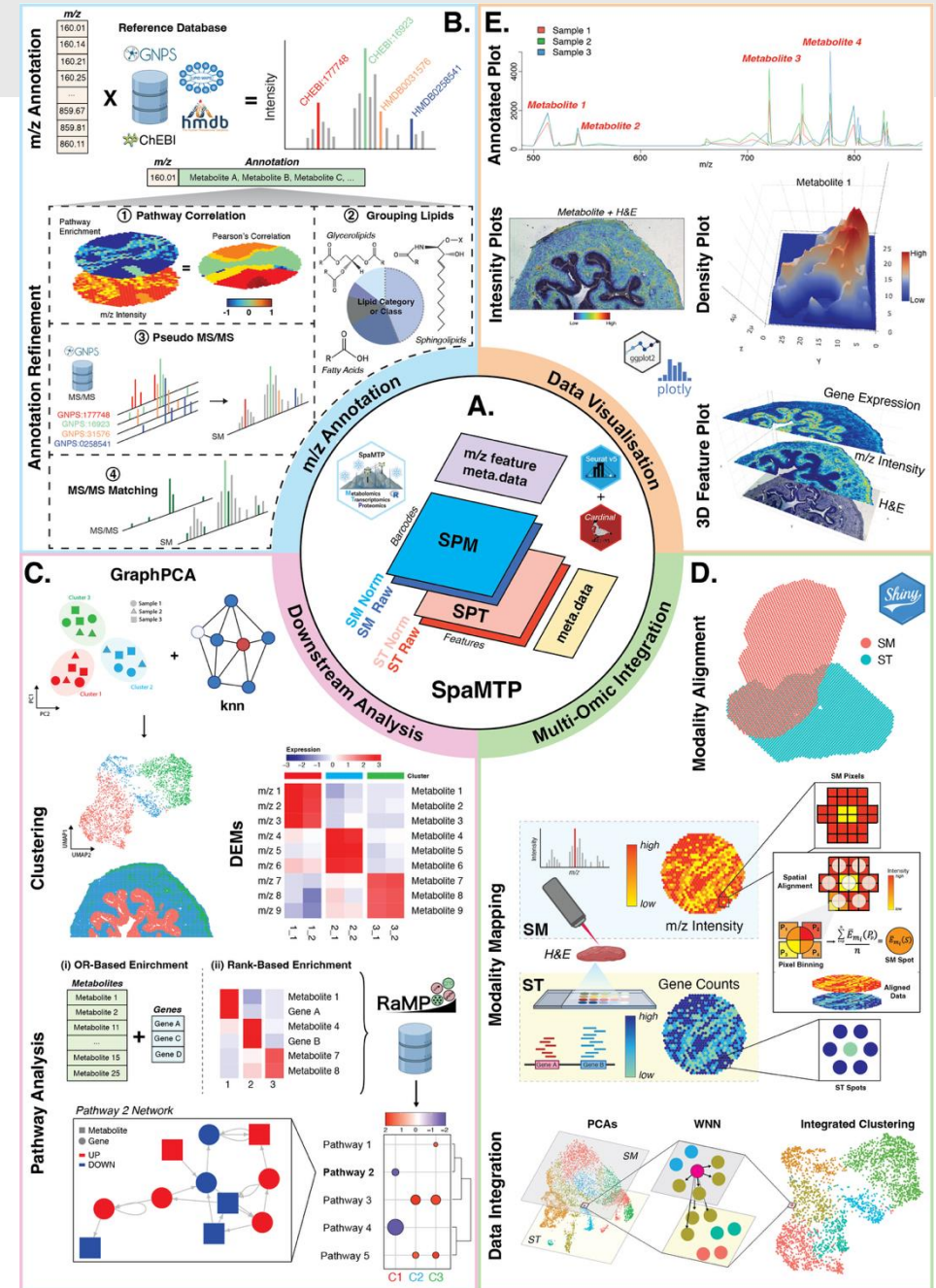
Squidpy V1.7.0 > Anndata

SpaGCN 3.0.0 > h5ad

Mantis

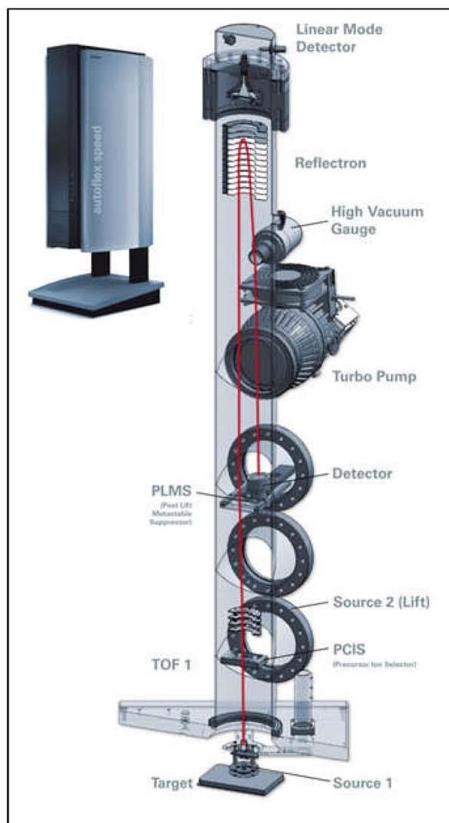
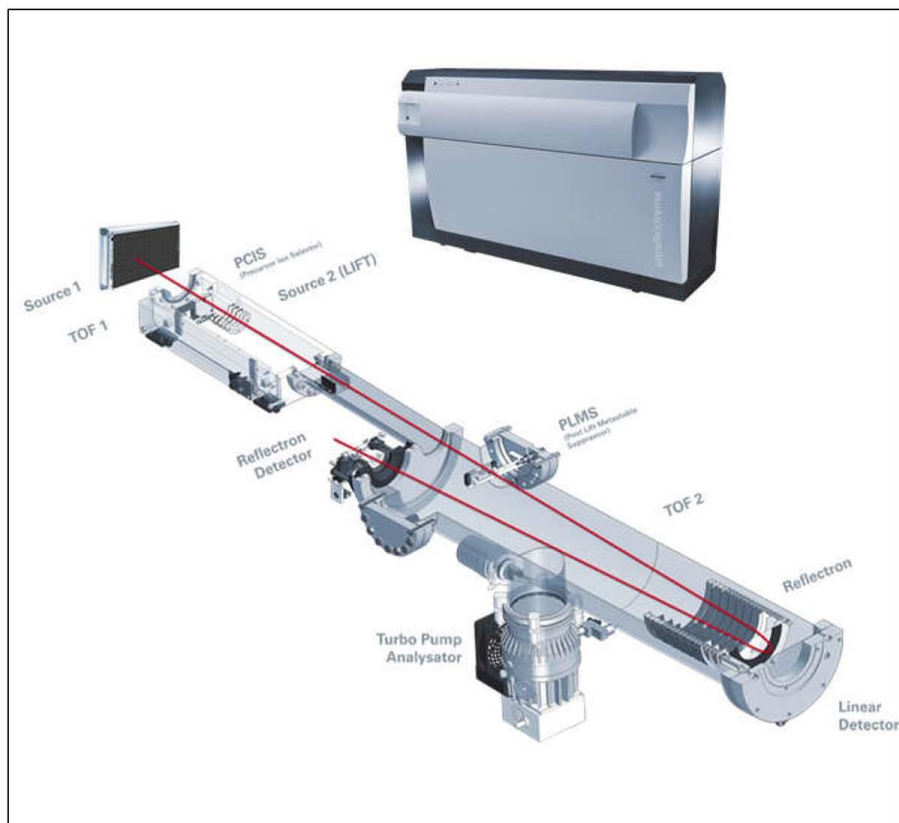
Magpie

SMIT

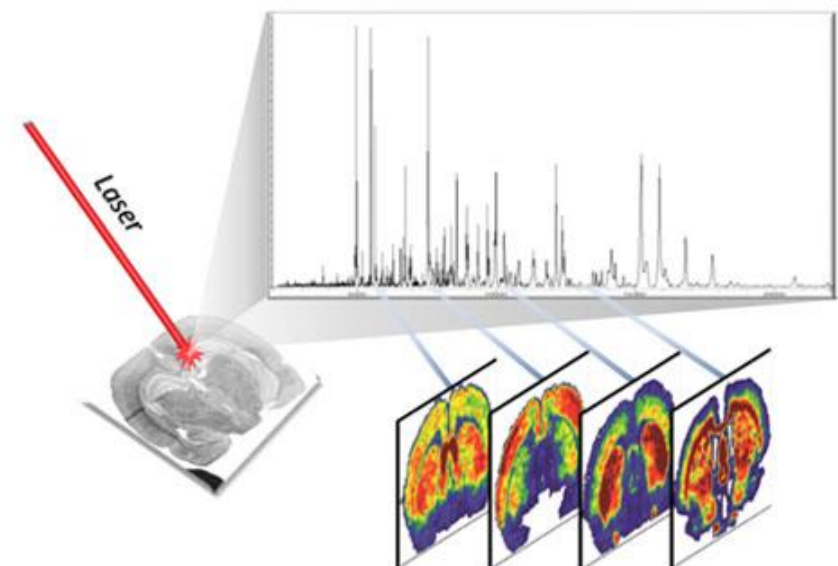


Spatial Metabolomics: MALDI

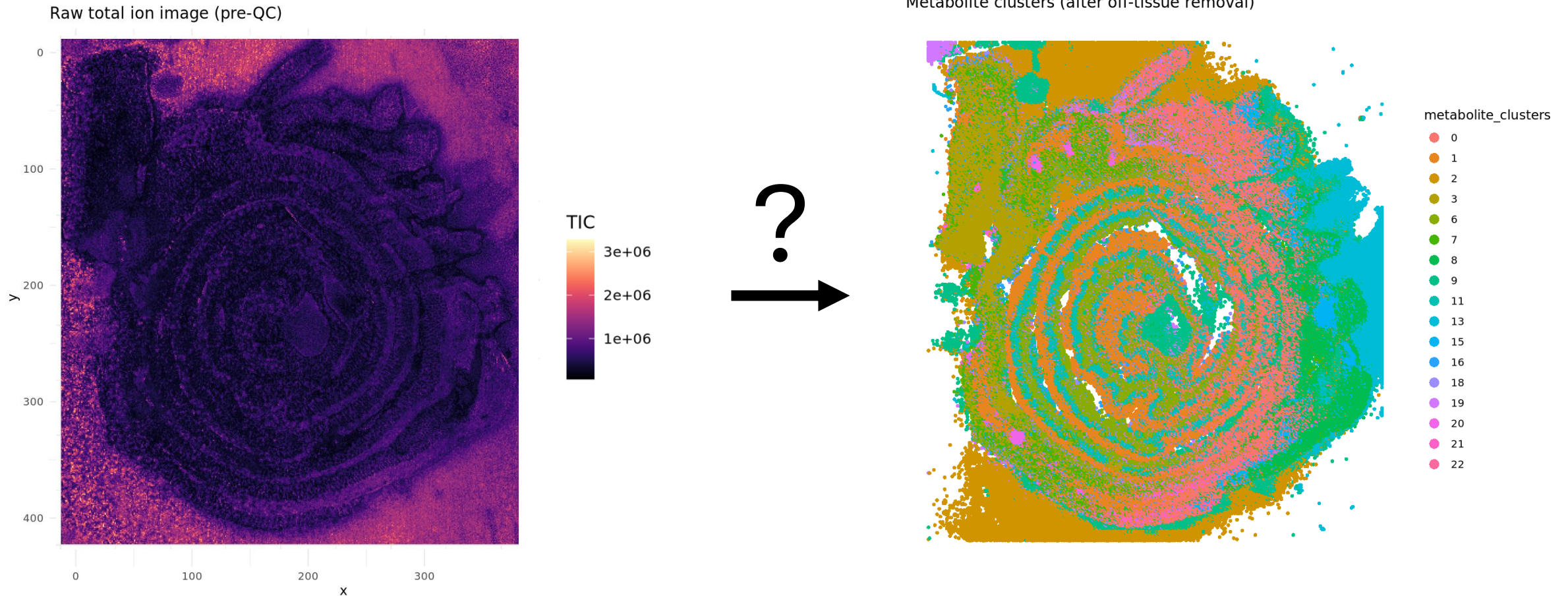
Diagrams of autoflex speed and ultrafleXtreme MALDI-TOF/TOF MS systems



Bruker Daltonics



Dataset analyzed with Cardinal and SpaMTP



First SM analysis steps need to be in HPC

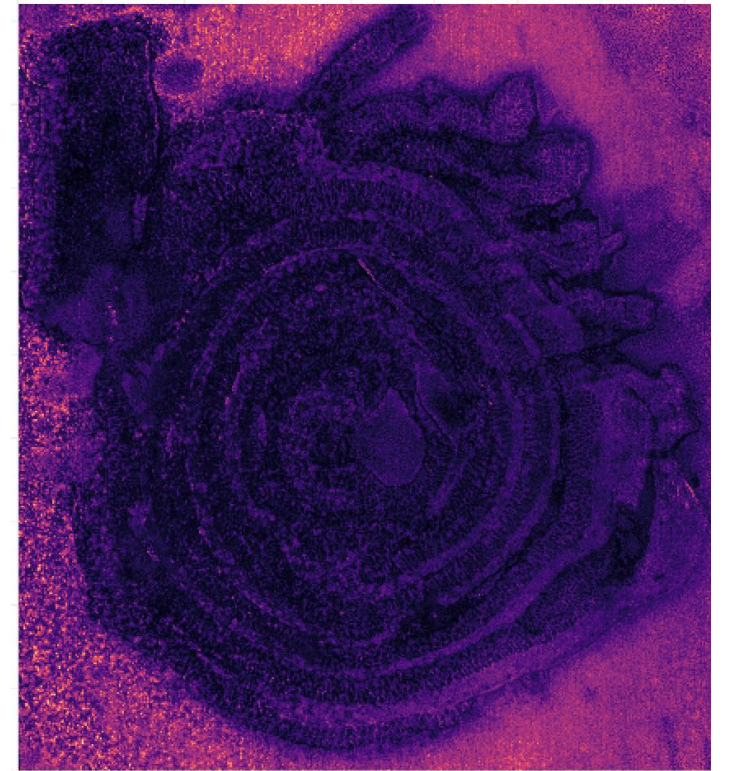
SM Dataset

Sample: wildtype mouse colon, fresh frozen in CMC, DAN matrix, negative ion mode

AP-SMALDI 10 / Orbitrap, 15 μm pixel size, m/z 200-1000

Raw data: 193,200 pixels (460 $\times$ 420), 46,447 detected m/z features

~2.5 GB `.ibd` + 288 MB `.imzML`



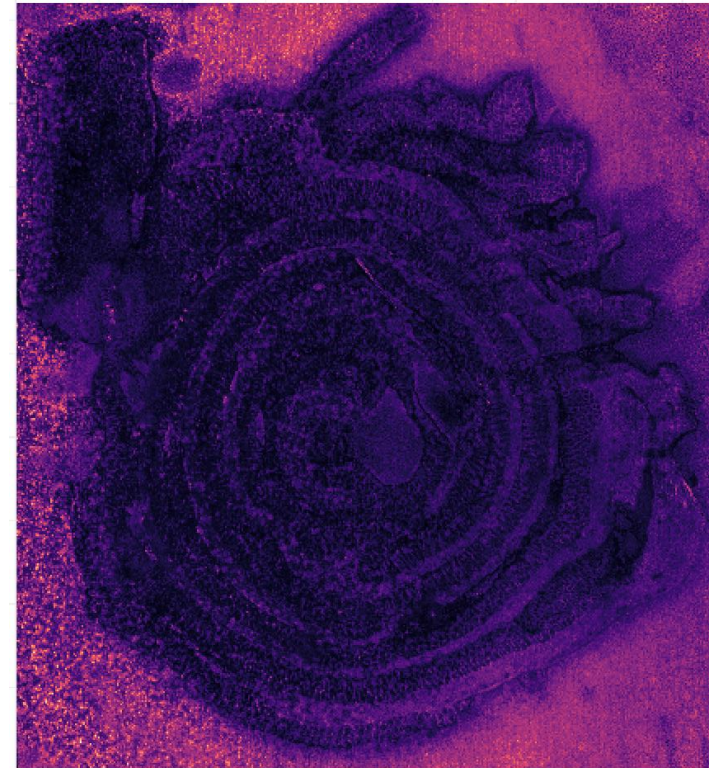
~9 billion data points before any reduction

First SM analysis steps need to be in HPC

1. Read the raw imzML/ibd pair with Cardinal: mass range 200-1000.

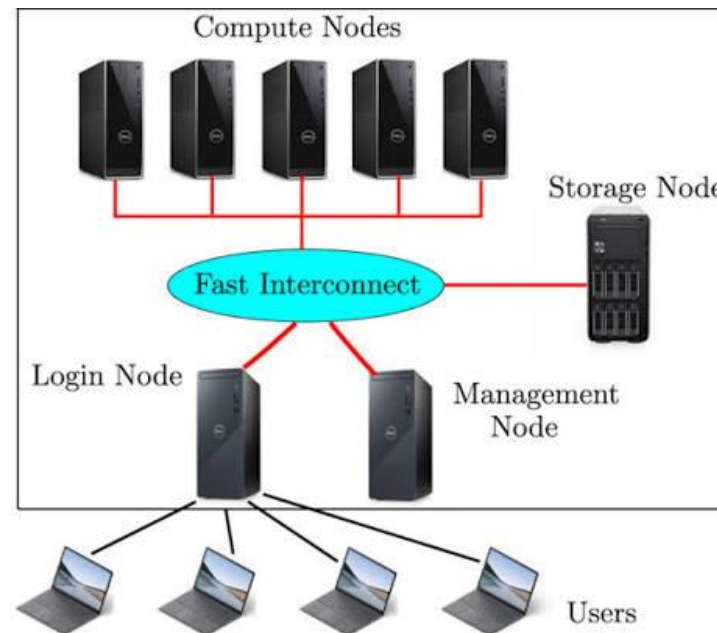
First SM analysis steps need to be in HPC

1. Read the raw imzML/ibd pair with Cardinal: mass range 200-1000.
2. Computed total ion current (TIC) per pixel and inspected it visually



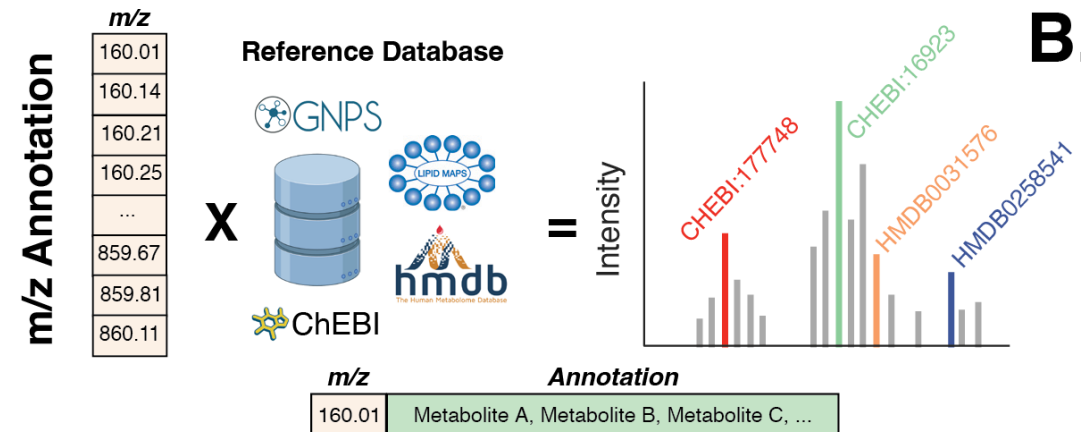
First SM analysis steps need to be in HPC

1. Read the raw imzML/ibd pair with Cardinal: mass range 200-1000.
2. Computed total ion current (TIC) per pixel and inspected it visually
3. *TIC-normalized: each spectrum, then filtered out very rare peaks (fewer than 1% of all pixels) likely to be noise or contamination*



First SM analysis steps need to be in HPC

1. Read the raw imzML/ibd pair with Cardinal: mass range 200-1000.
2. Computed total ion current (TIC) per pixel and inspected it visually
3. *TIC-normalized: each spectrum, then filtered out very rare peaks (fewer than 1% of all pixels) likely to be noise or contamination*
4. *Converted to a Seurat-style object as the transcriptomics side of the course.*
5. *Annotated every retained m/z feature against three reference databases (HMDB, LipidMaps, ChEBI)*
6. *Saved this as an **annotated object***



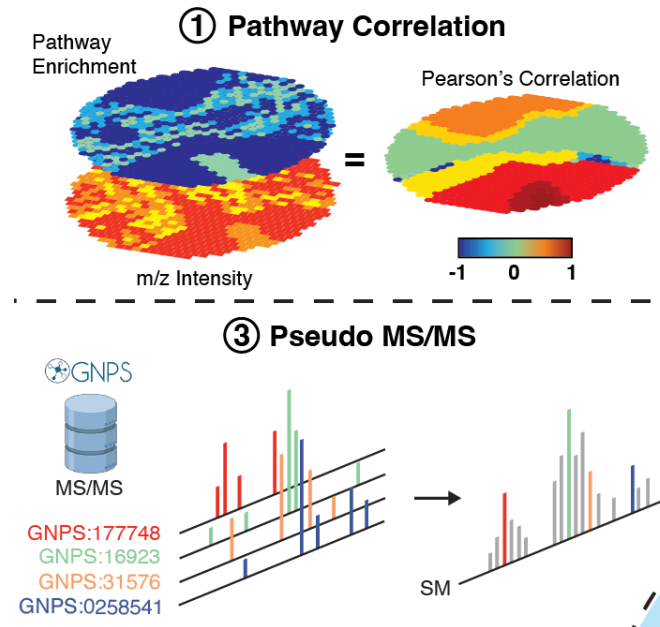
Continue the analysis from full object

Subset the SM object for the integration

From 193,200 pixels × thousands of features.

Known metabolites only

3 or fewer candidate compound names annotations specific



Subset the SM object for the integration

From 193,200 pixels × thousands of features.

Known metabolites only

3 or fewer candidate compound names annotations specific

Off-tissue background removed

Comparing those clusters against the TIC image: clusters had distinctly higher mean TIC than the rest (bare MALDI matrix).

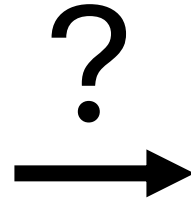
SM analyzed with Carousel and SpaMTP

Metabolite clusters (full dataset)



metabolite_clusters

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16
- 17



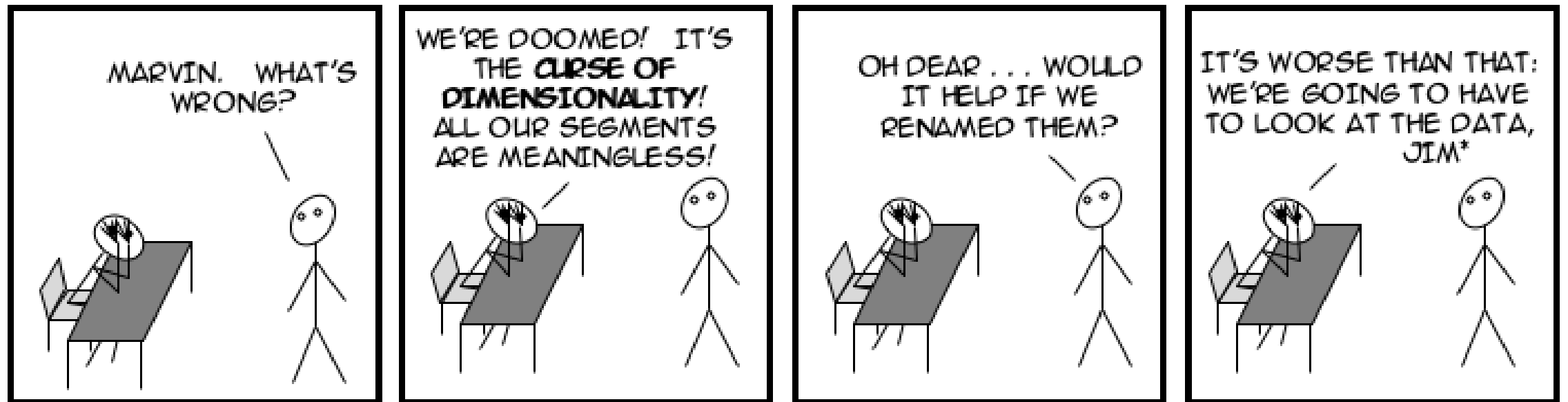
Off-tissue test: clusters 12, 17, 4, 5, 14, 10



off_tissue_test

- Would remove
- Would keep

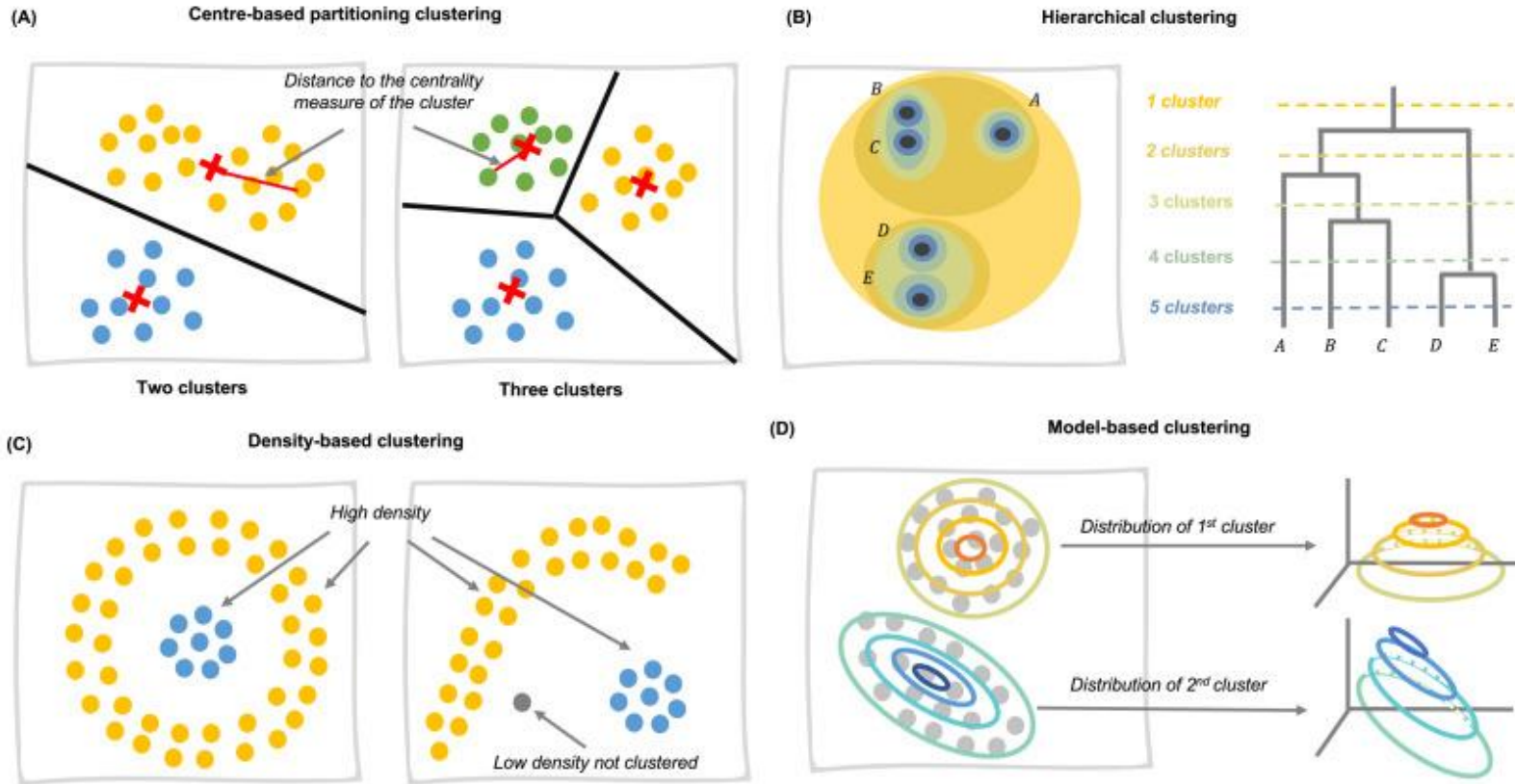
What is clustering?



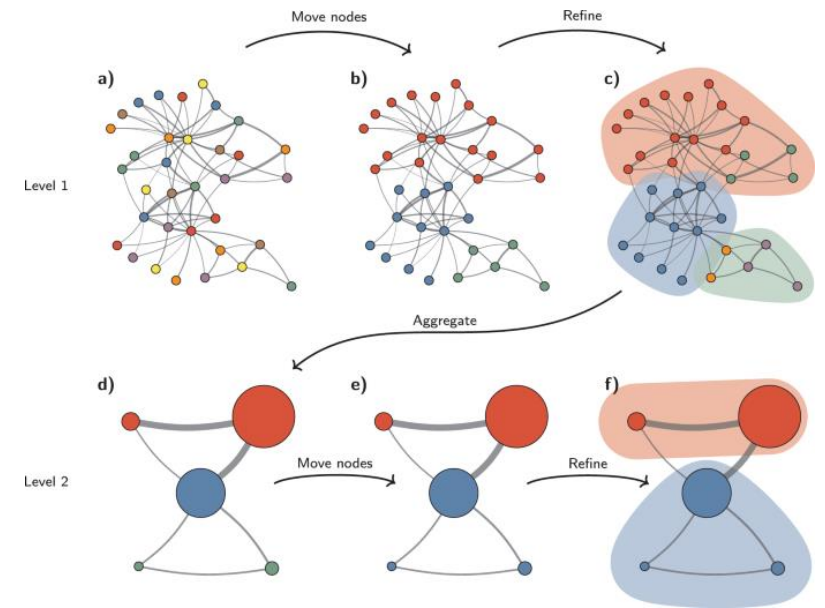
[HTTP://SCIENTIFICMARKETER.COM](http://scientificmarketer.com)

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* WITH APOLOGIES TO MR SPOCK & STAR TREK.

What is clustering?



Graph-based Leiden clustering



Caroline X. Gao An overview of clustering methods with guidelines for application in mental health research, *Psychiatry Research*, Volume 327, 2023, <https://doi.org/10.1016/j.psychres.2023.115265>.

Traag, V.A., Waltman, L. & van Eck, N.J. From Louvain to Leiden: guaranteeing well-connected communities. *Sci Rep* **9**, 5233 (2019). <https://doi.org/10.1038/s41598-019-41695-z>

Subset the SM object for the integration

From 193,200 pixels × thousands of features.

Known metabolites only

3 or fewer candidate compound names annotations specific

Off-tissue background removed

Comparing those clusters against the TIC image: clusters had distinctly higher mean TIC than the rest (bare MALDI matrix).

A curated, representative feature panel

Statistically distinct markers: for each tissue region

Spatially patterned features: vary meaningfully across the tissue

Class SM dataset

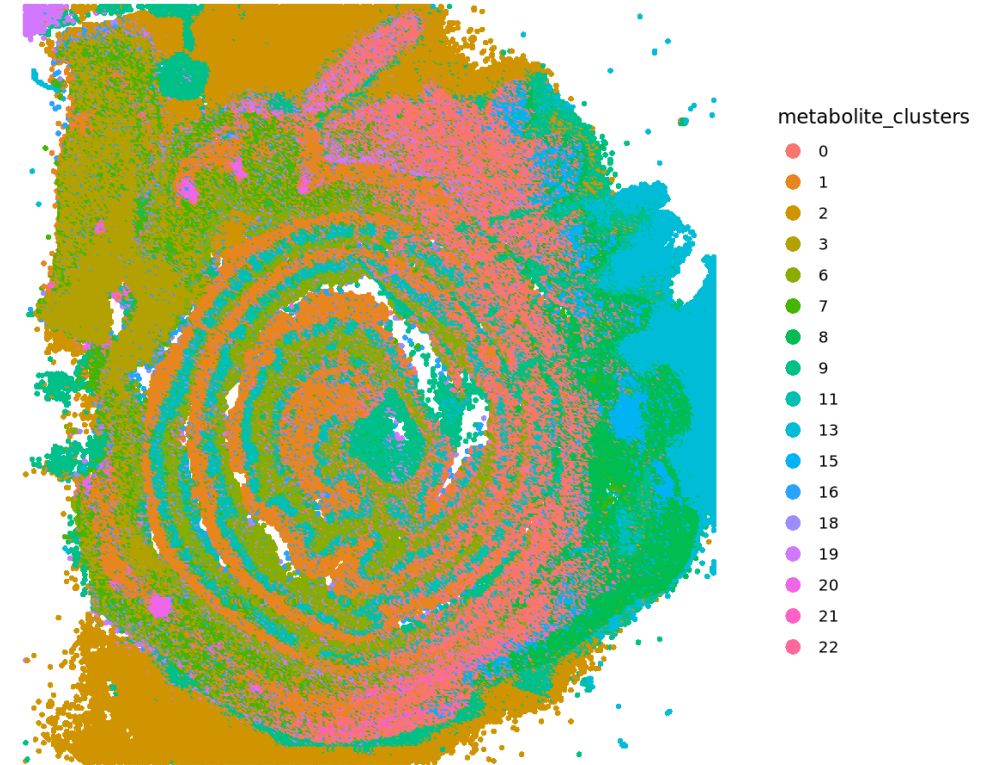
Sample: wildtype mouse colon, fresh frozen in CMC, DAN matrix, negative ion mode

125,533 pixels

869 features

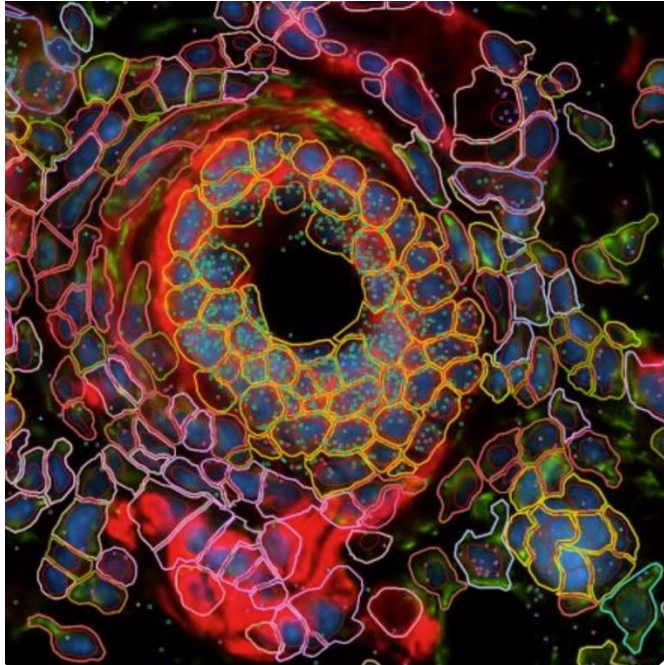
221.4 MB

Metabolite clusters (after off-tissue removal)



Spatial Transcriptomics: Two different technologies

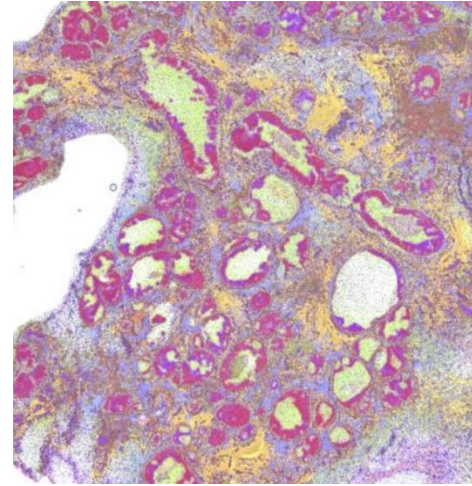
In situ hybridization



Up to 5000 RNAs & 100 proteins

e.g. 10x Xenium

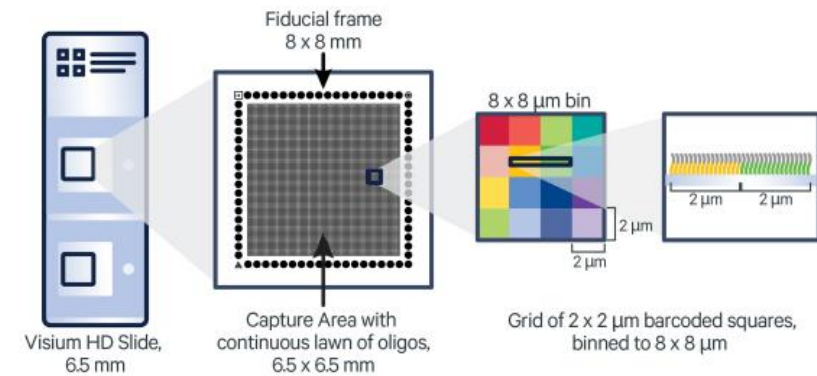
Unbiased whole transcriptome



**Probe-based solutions
18,000 genes**

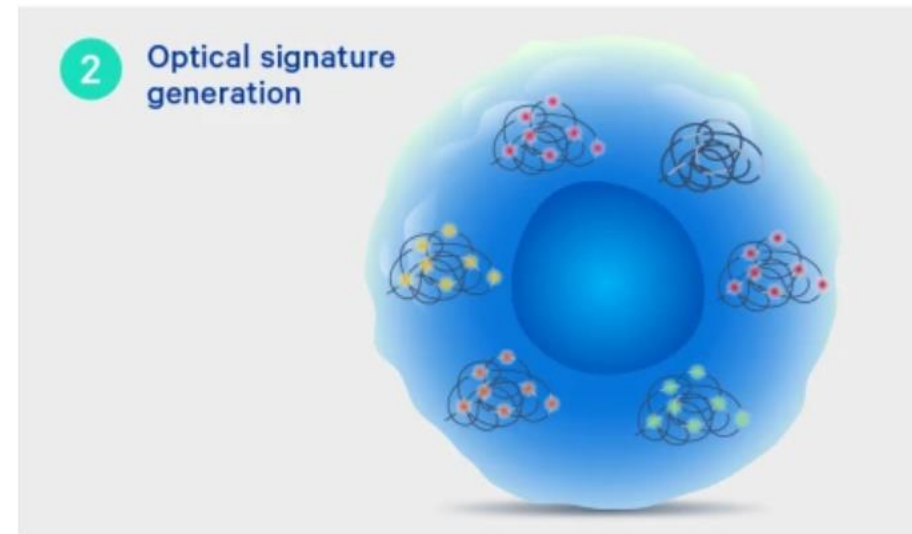
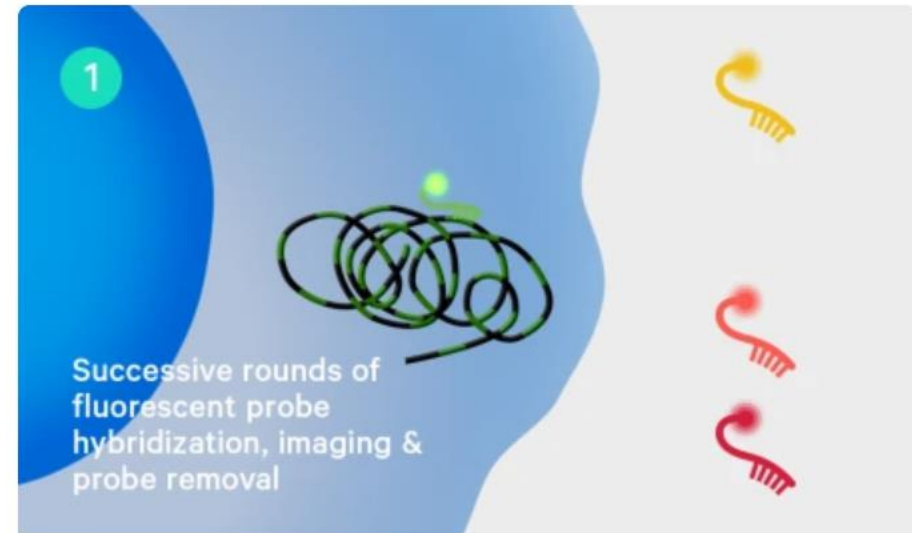
or

3' Poly-A capture



e.g. 10x Visium HD

10x Xenium



10x Xenium



Steps 1 & 2: Dual target recognition & hybridization

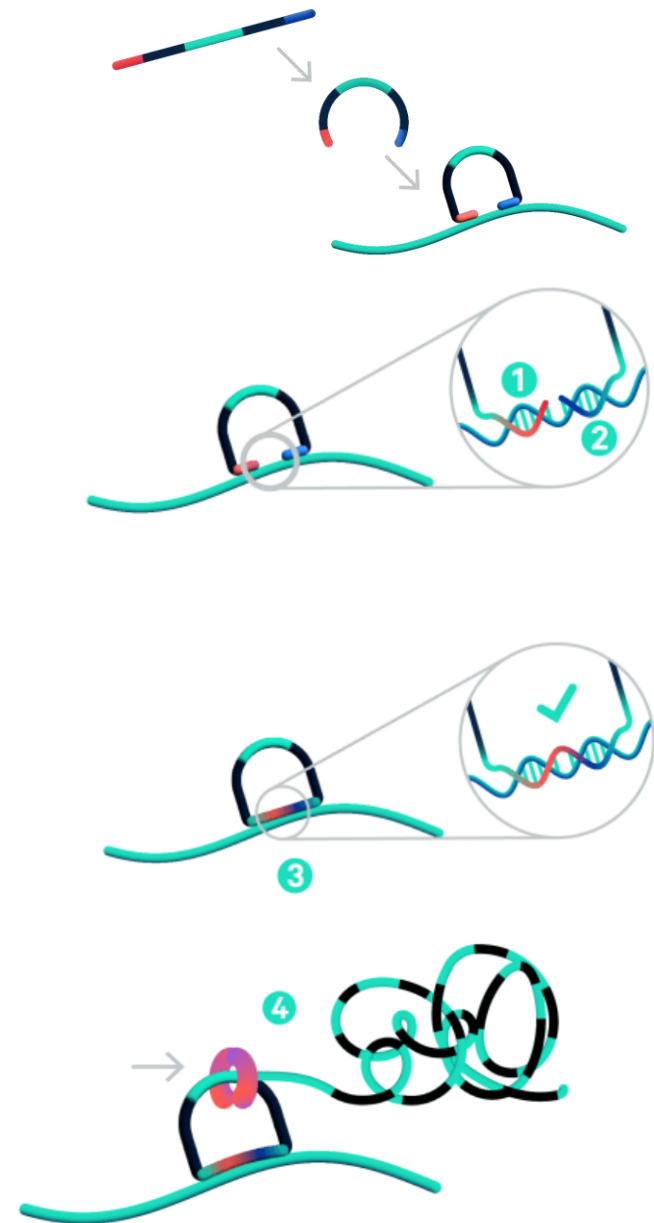
Both probe arms must stably hybridize to their target. If only one arm hybridizes, the probe is unstable and will be washed off in the post hybridization wash. The padlock probes also contain a gene-specific barcode sequence that is used to generate a unique optical signature of each transcript.

Step 3: Probe ligation

Only when both probe arms are stably hybridized to the target with a perfect match does proximal probe ligation occur. Partially bound or mis-matching probes are not ligated and cannot be amplified.

Step 4: Rolling circle amplification (RCA)

Only ligated probes are amplified, enriching for true target recognition events and suppressing off-target events. The RCA approach creates a robust and strong signal allowing single probe detection.



Spatial Biology: Two different technologies

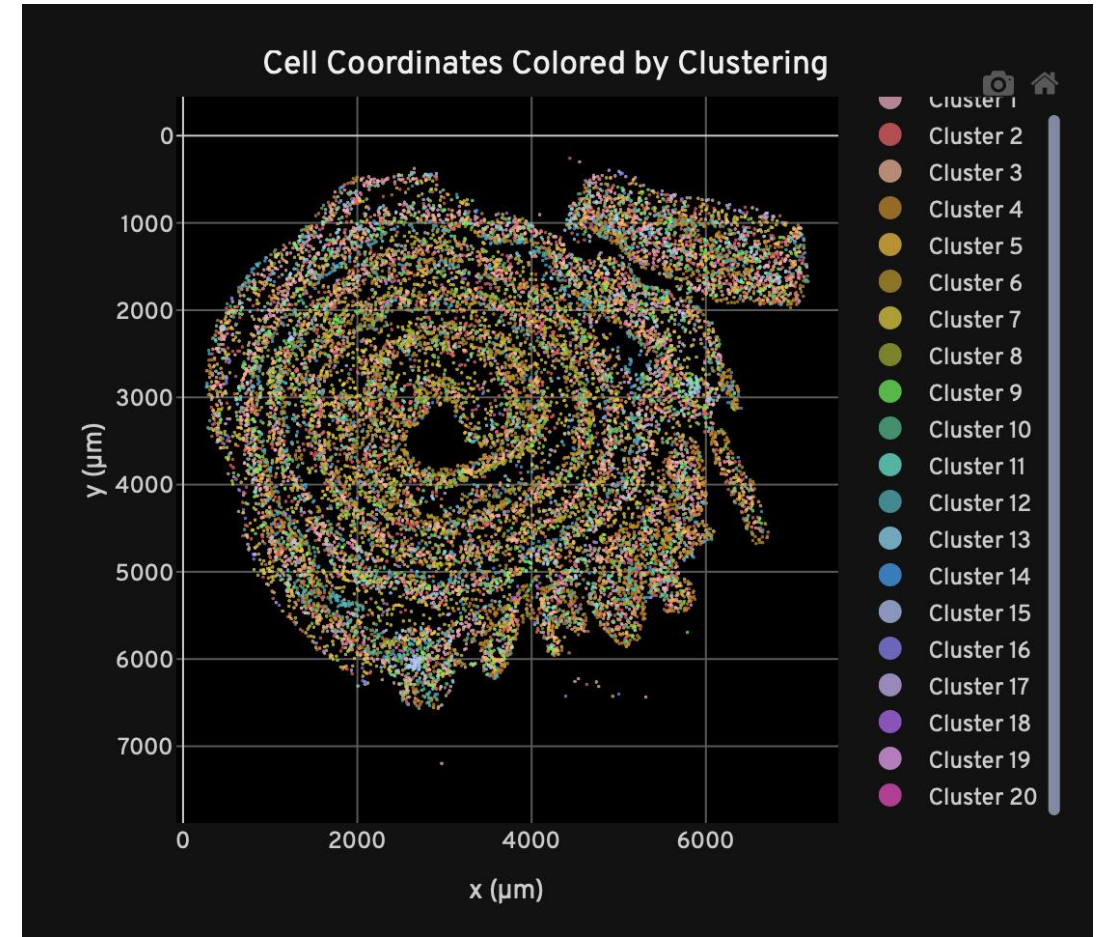
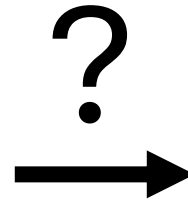
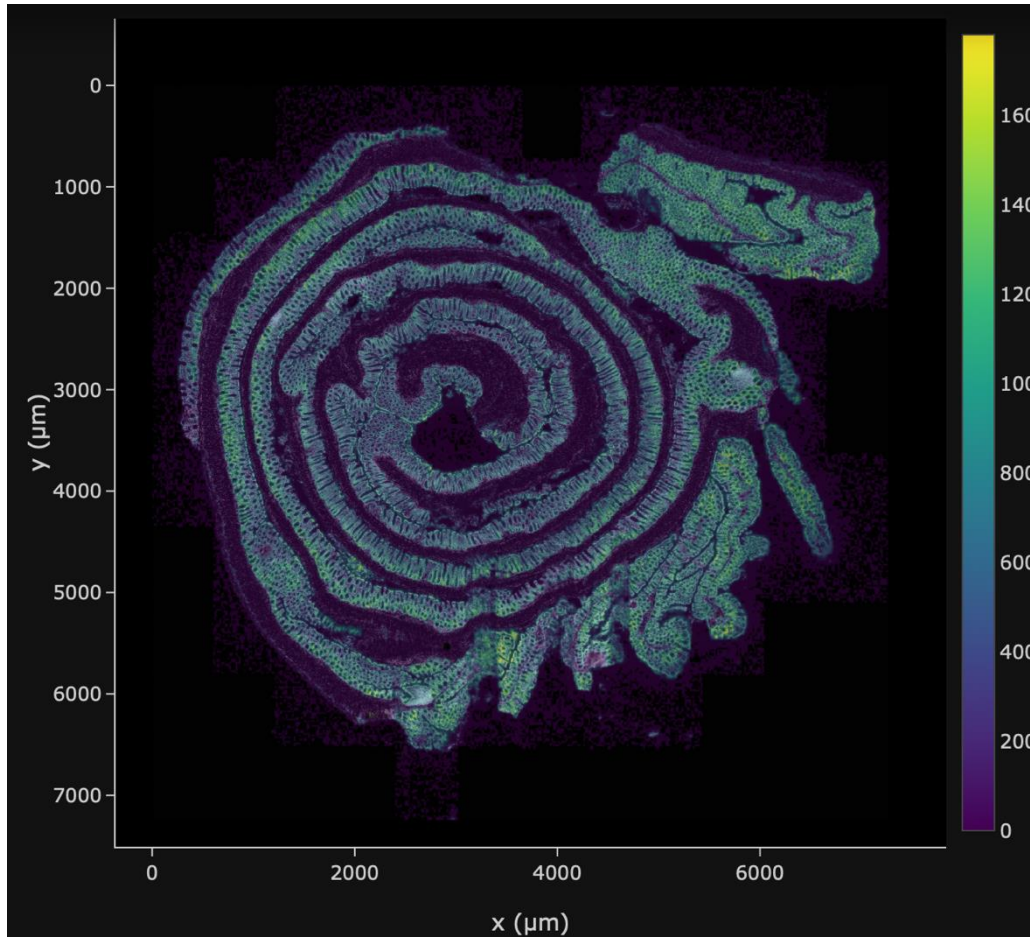
10x Xenium



Up to 5000 RNAs & 100 proteins

Costume panel of 100 features: Metabolism

ST analyzed with Seurat and SpaMTP



First ST analysis steps need to be in HPC

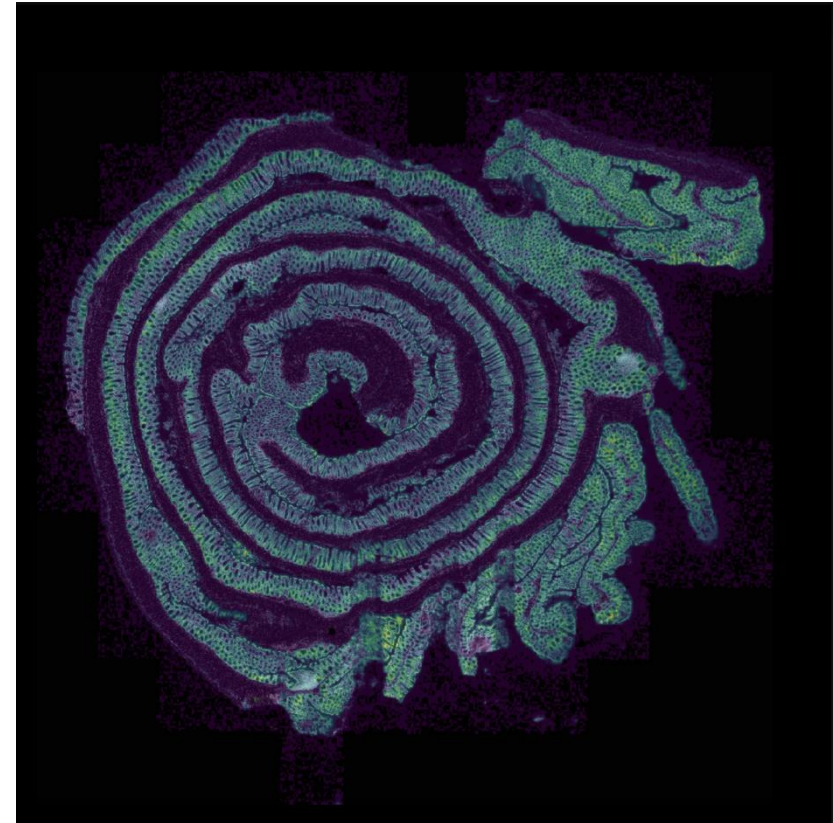
ST Dataset

Sample: wildtype mouse colon, fresh frozen in CMC

Instrument: Xenium v1

Raw data: 317,592 number of cells detected, 60 transcripts per cell

1 cut: 12.3Gb



~32 million data points before any reduction

Subset the ST object for the integration

From 317,592 cells × 100 features.

Only meaningful cells

Keep cells with 5 or more genes (100 gene panel)

Only high-quality cells

Keep cells with more than 10 transcripts

237,669 cells x 100 features: 78.8 MB

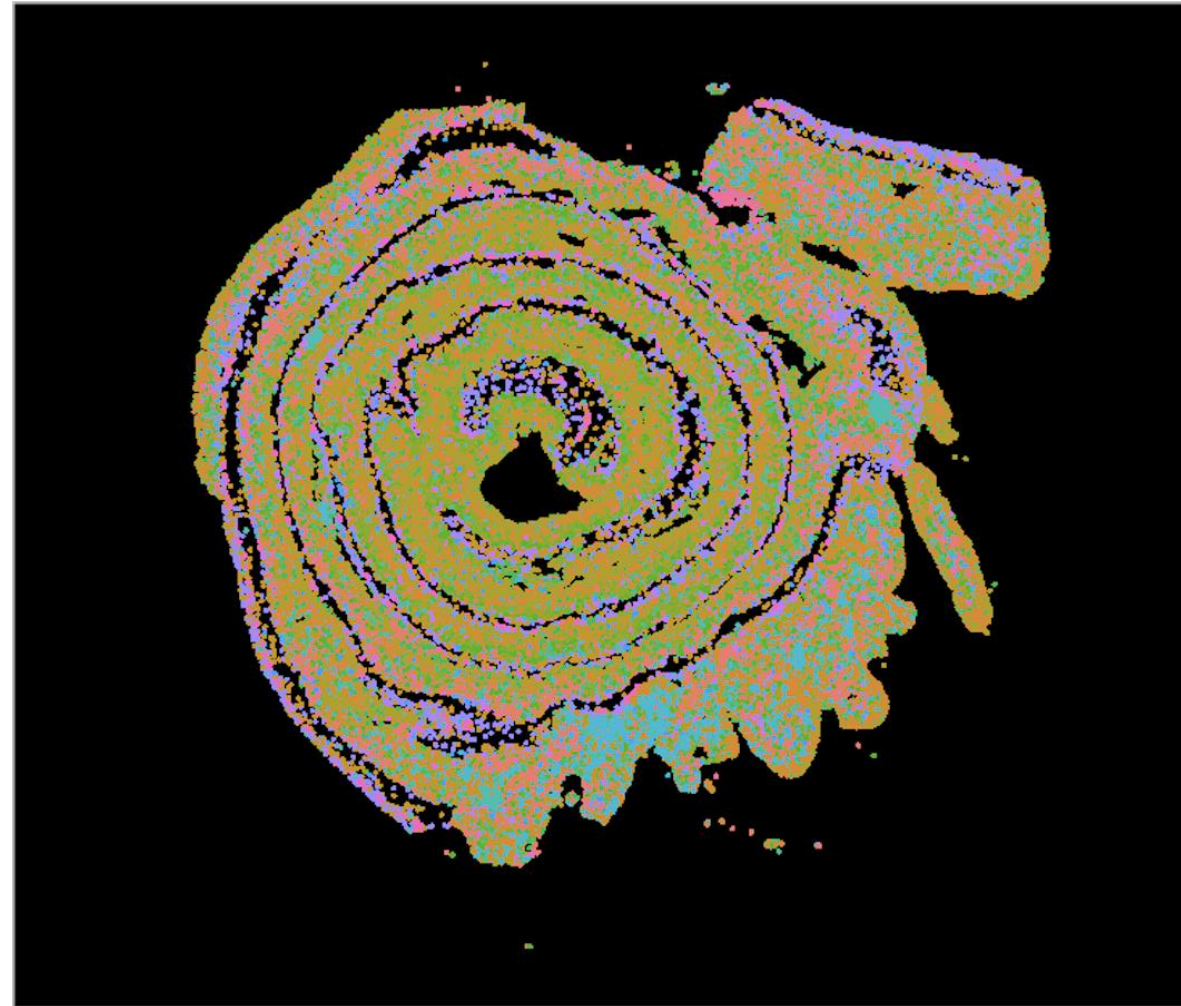
Class ST dataset

Sample: wildtype mouse colon,
fresh frozen in CMC

237,669 cells

100 features

78.8 MB



Getting access to your code

- Please go to <https://github.com/Kiel-Spatial-Biology-Center>
- Retrieve your R markdown code
- Link to Dataset
- Link to additional Datasets of interest

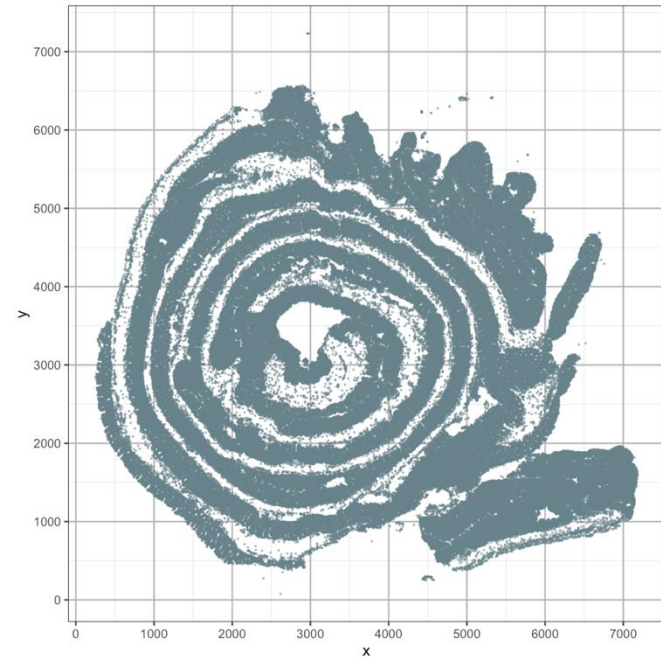


SpaMTP: 03_integration_spamtp.R

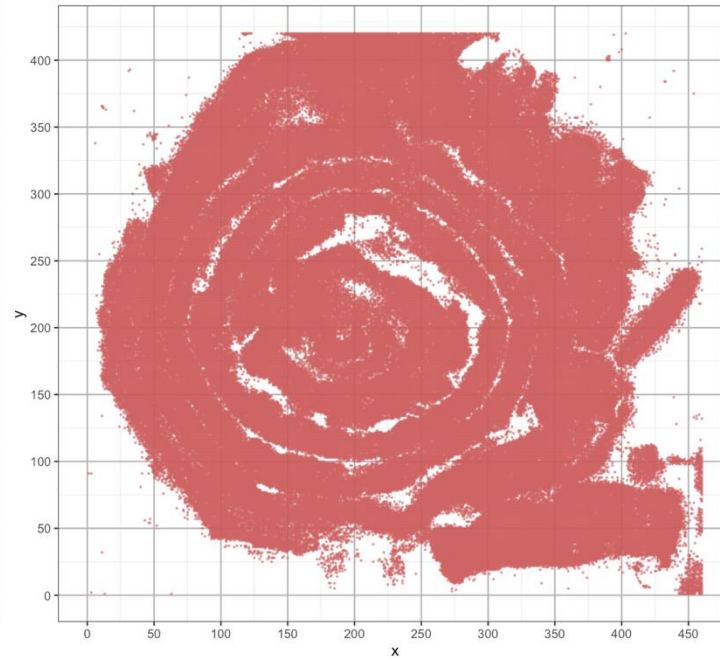
1. Load SM and ST
2. Align SM and ST
3. Map metabolites onto cells (transcriptomics)
4. (Cluster SM and ST separately)
5. Integrate dataset and multimodal clustering

2. Align SM and ST

Xenium coordinates (pick landmark x,y from here)



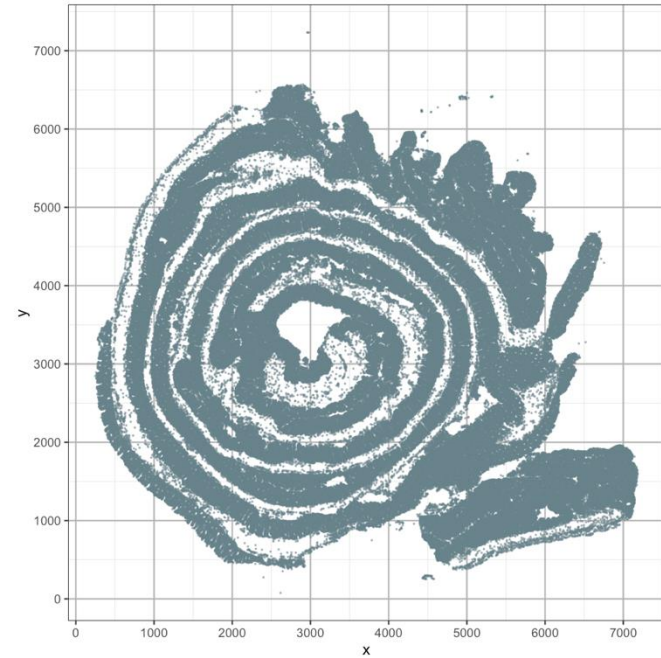
MALDI coordinates (pick landmark x,y from here)



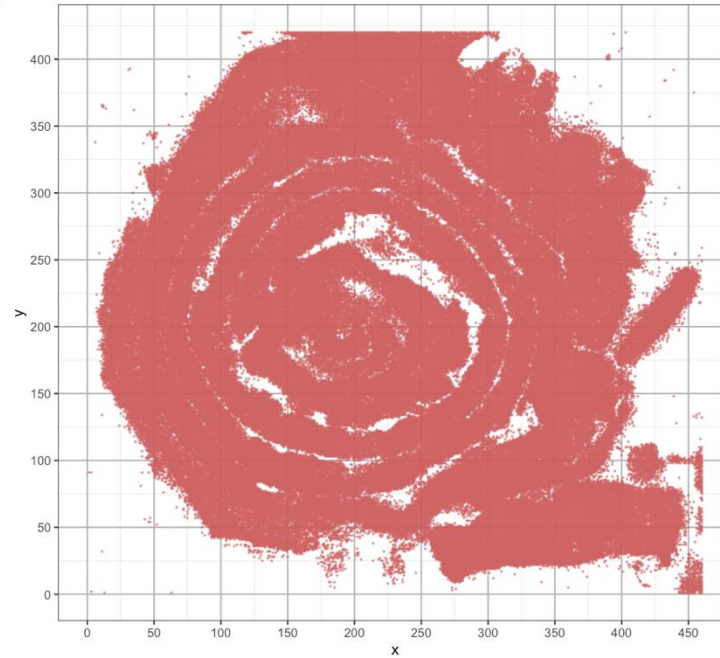
Why are they different?

2. Align SM and ST

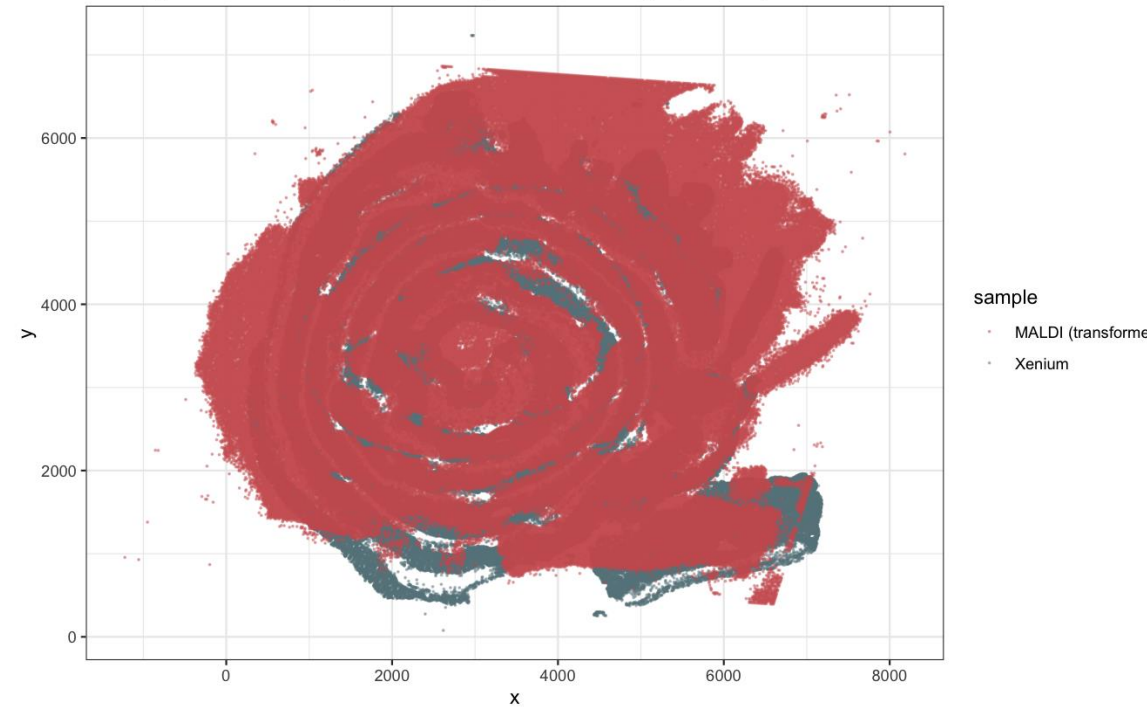
Xenium coordinates (pick landmark x,y from here)



MALDI coordinates (pick landmark x,y from here)



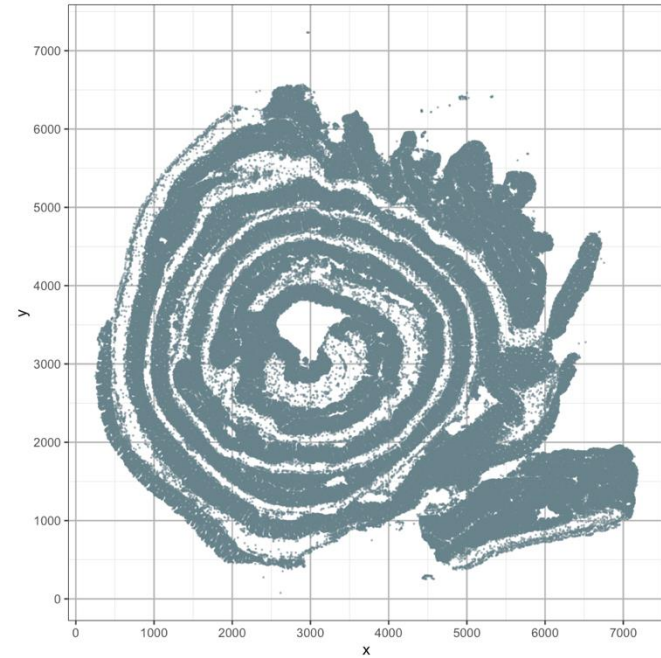
Overlay after manual alignment -- shapes should roughly correspond



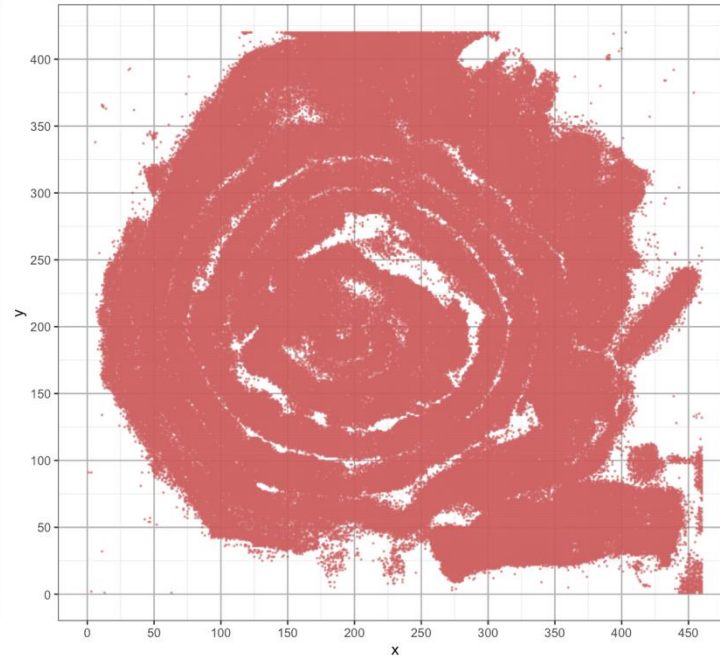
Well aligned?

2. Align SM and ST

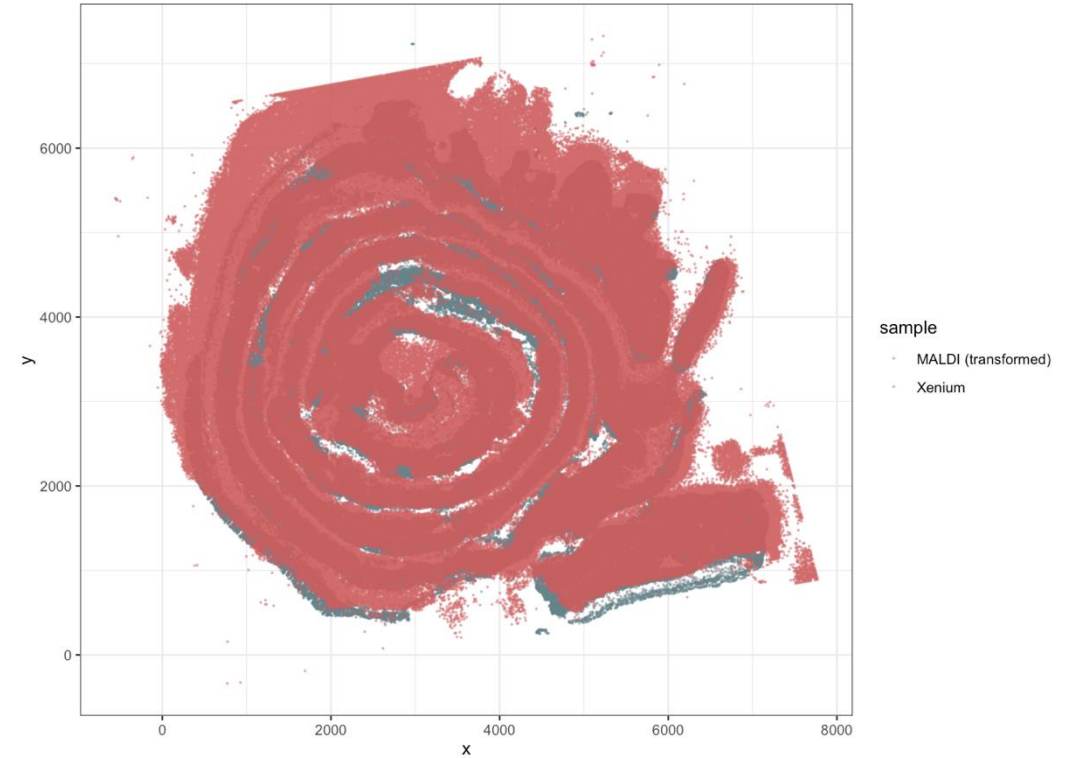
Xenium coordinates (pick landmark x,y from here)



MALDI coordinates (pick landmark x,y from here)



Overlay after manual alignment -- shapes should roughly correspond



Better...

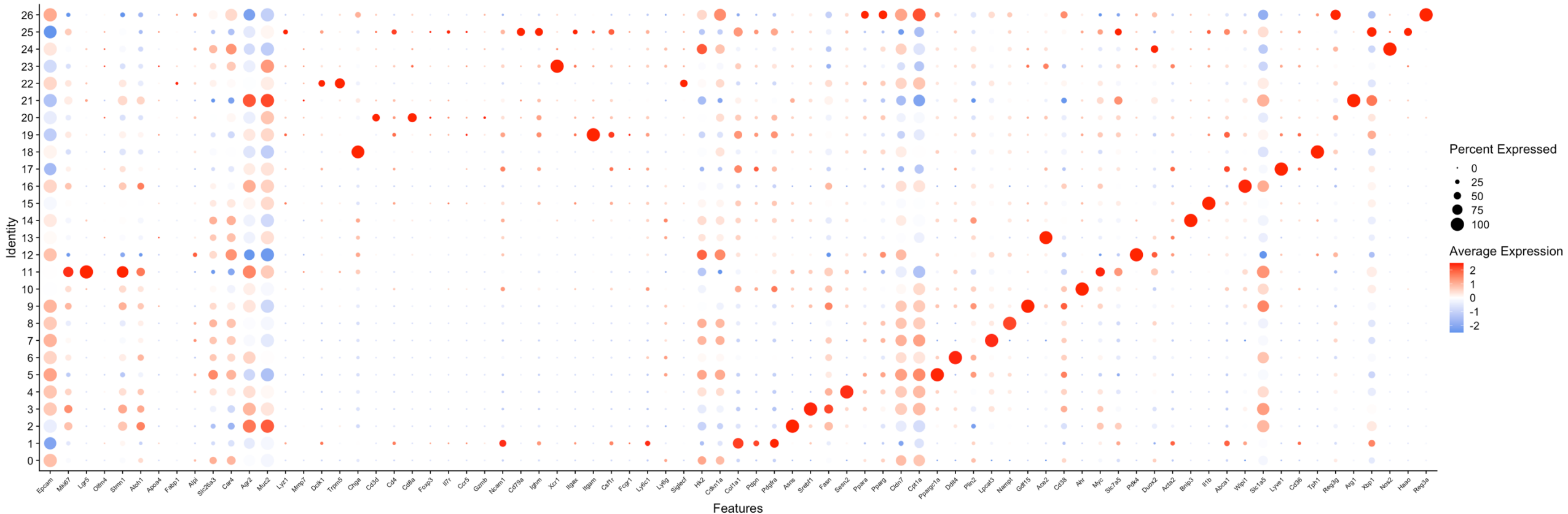
3. Map metabolites to cells

- First cell-type annotation base on ST

For more information please see

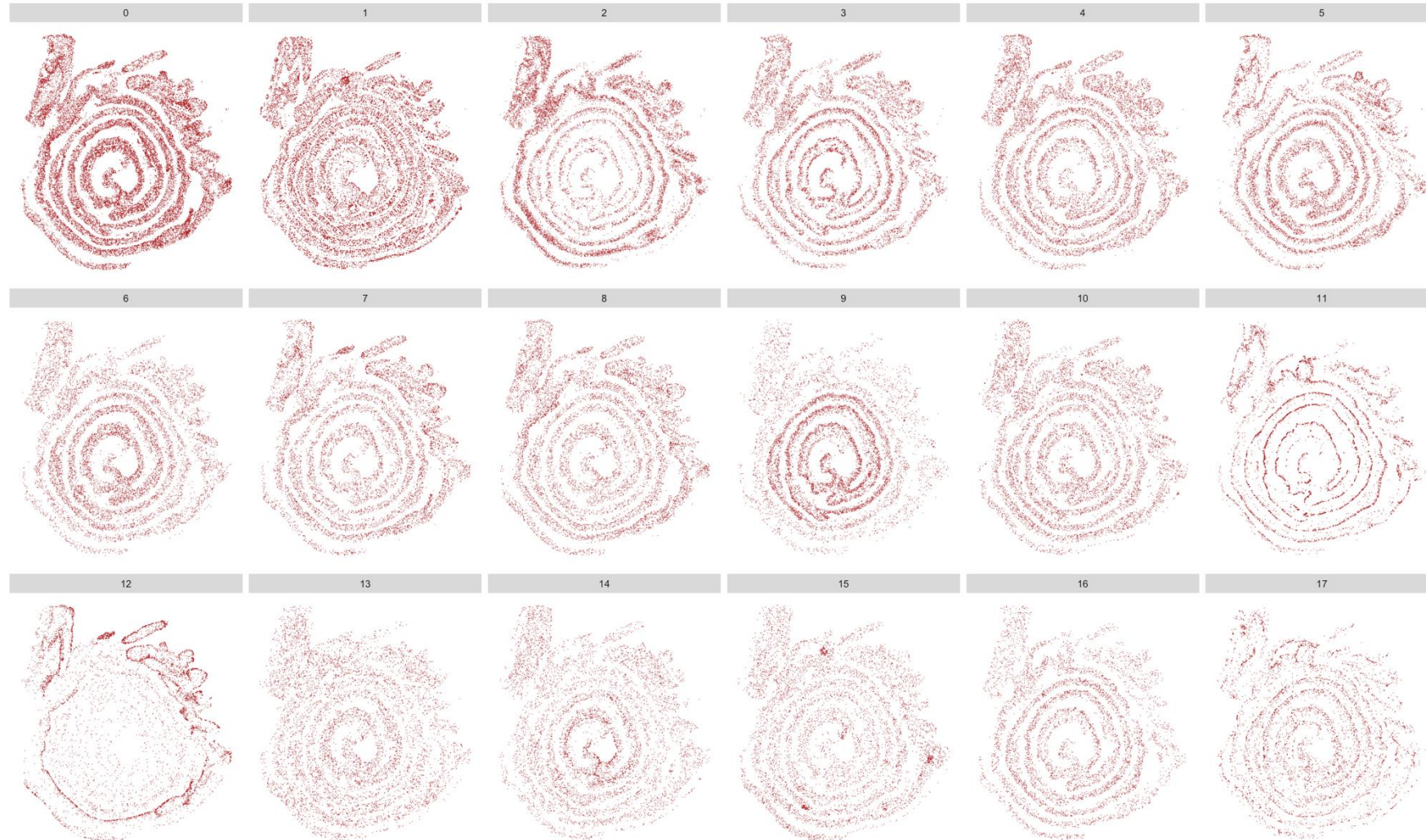
https://github.com/Kiel-Spatial-Biology-Center/spatialRNAseq_workshop25

4. Celltype calling: Cluster markers

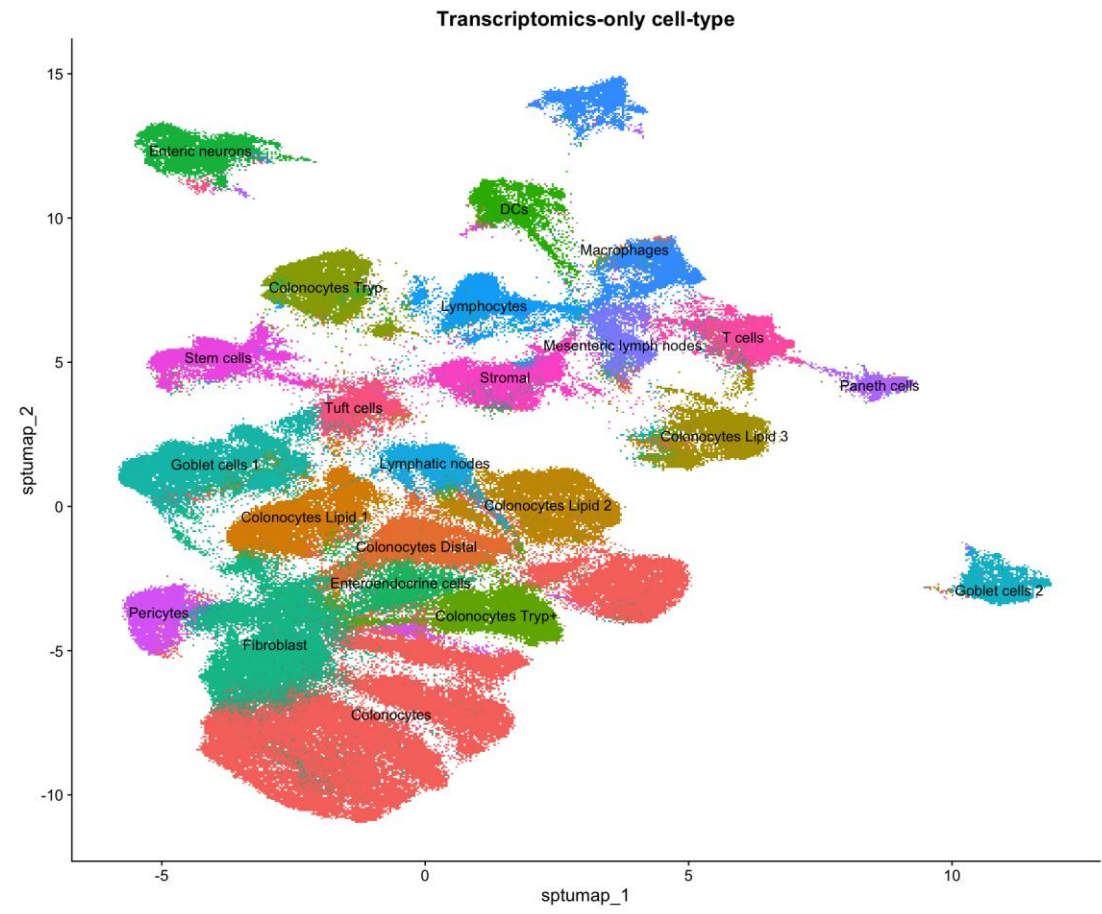


4. Celltype calling: Cluster in tissue

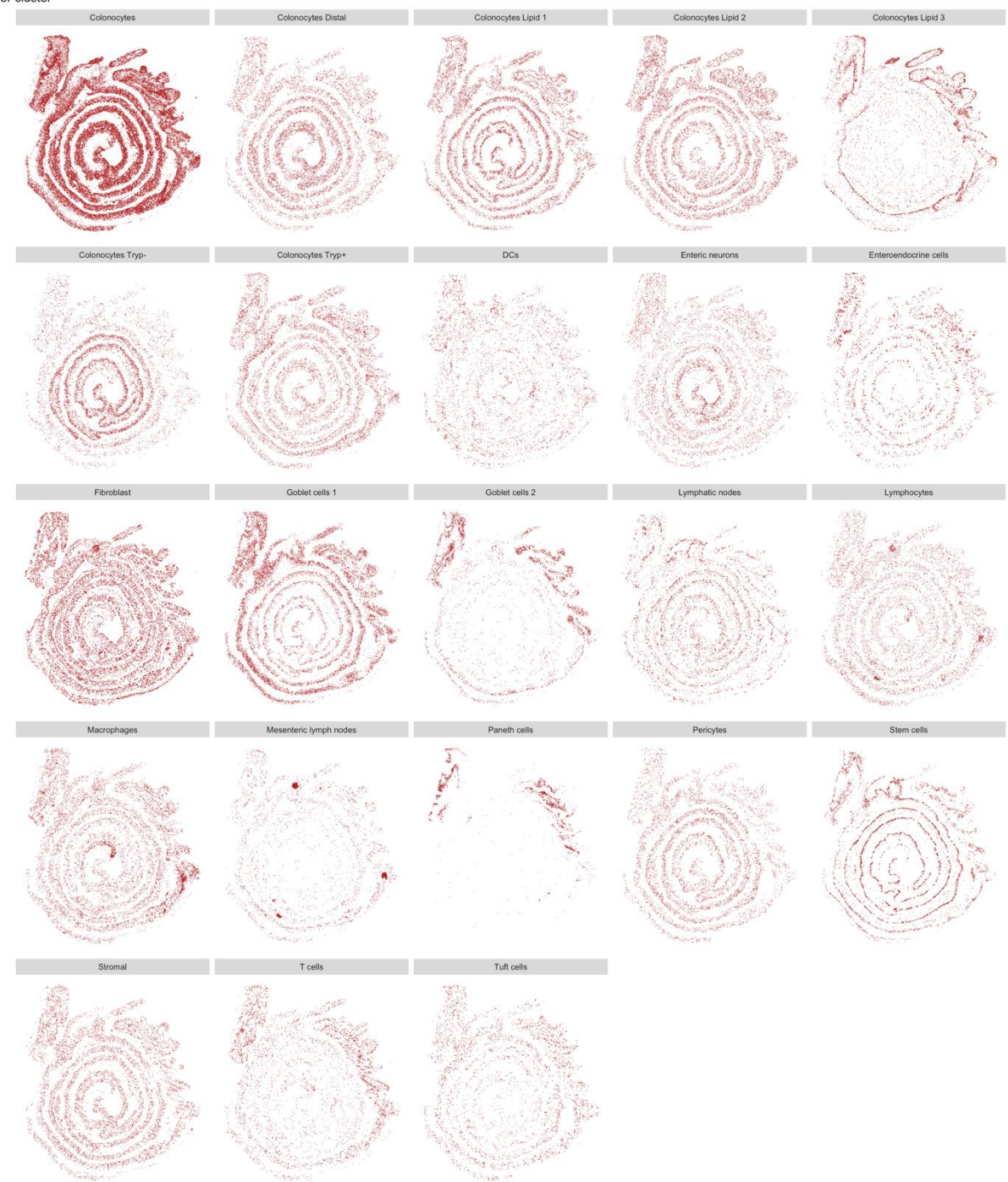
ST_clusters -- spatial location per cluster



4. Cell type calling

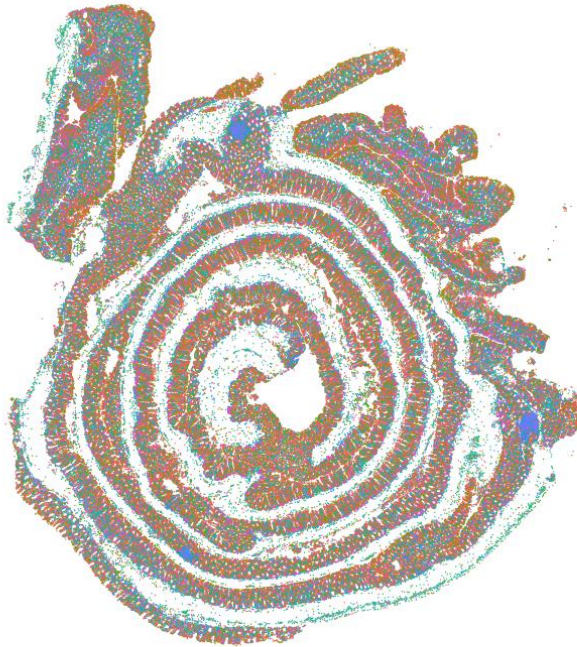


cell_type -- spatial location per cluster



4. Cell type calling

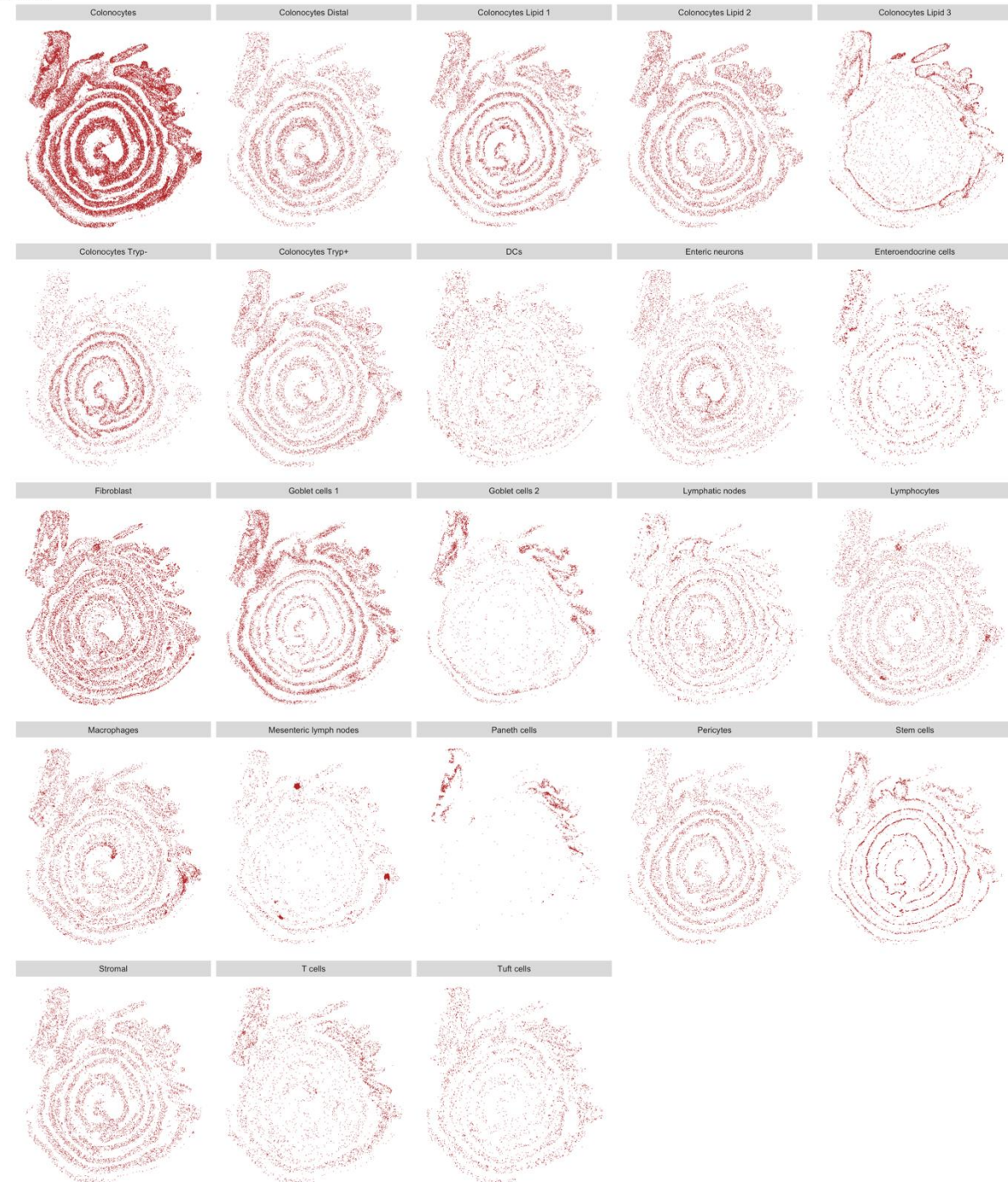
Cell-type annotation, spatial



cell_type

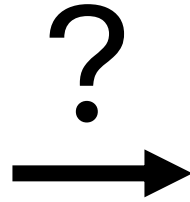
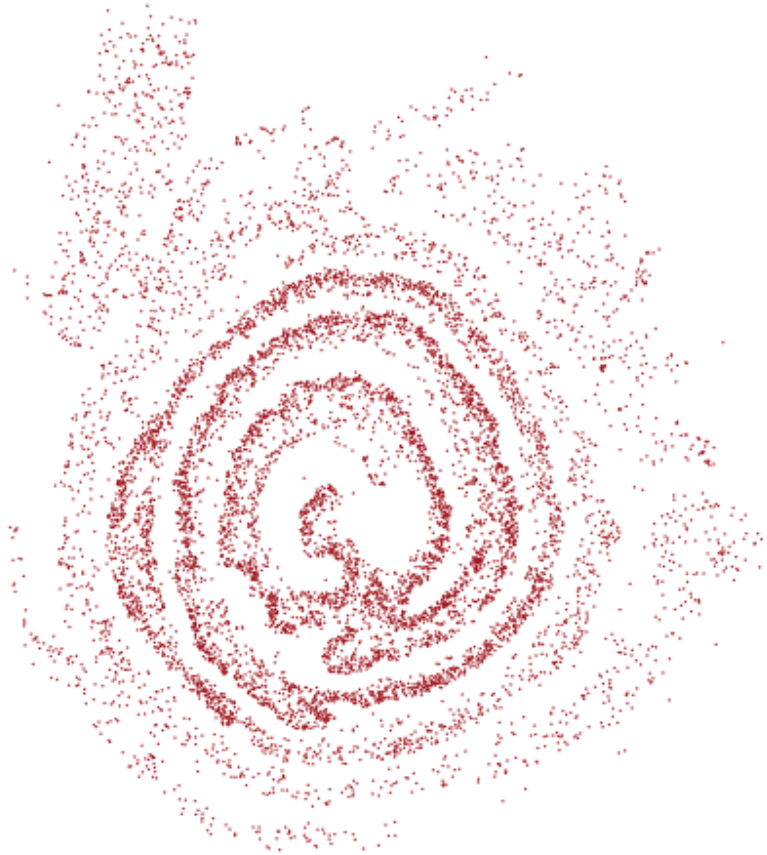
- | | |
|-------------------------|--------------------------|
| ● Colonocytes | ● Goblet cells 2 |
| ● Colonocytes Distal | ● Lymphatic nodes |
| ● Colonocytes Lipid 1 | ● Lymphocytes |
| ● Colonocytes Lipid 2 | ● Macrophages |
| ● Colonocytes Lipid 3 | ● Mesenteric lymph nodes |
| ● Colonocytes Tryp- | ● Paneth cells |
| ● Colonocytes Tryp+ | ● Pericytes |
| ● DCs | ● Stem cells |
| ● Enteric neurons | ● Stromal |
| ● Enteroendocrine cells | ● T cells |
| ● Fibroblast | ● Tuft cells |
| ● Goblet cells 1 | |

cell_type -- spatial location per cluster



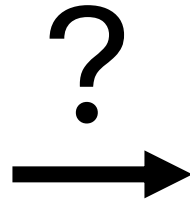
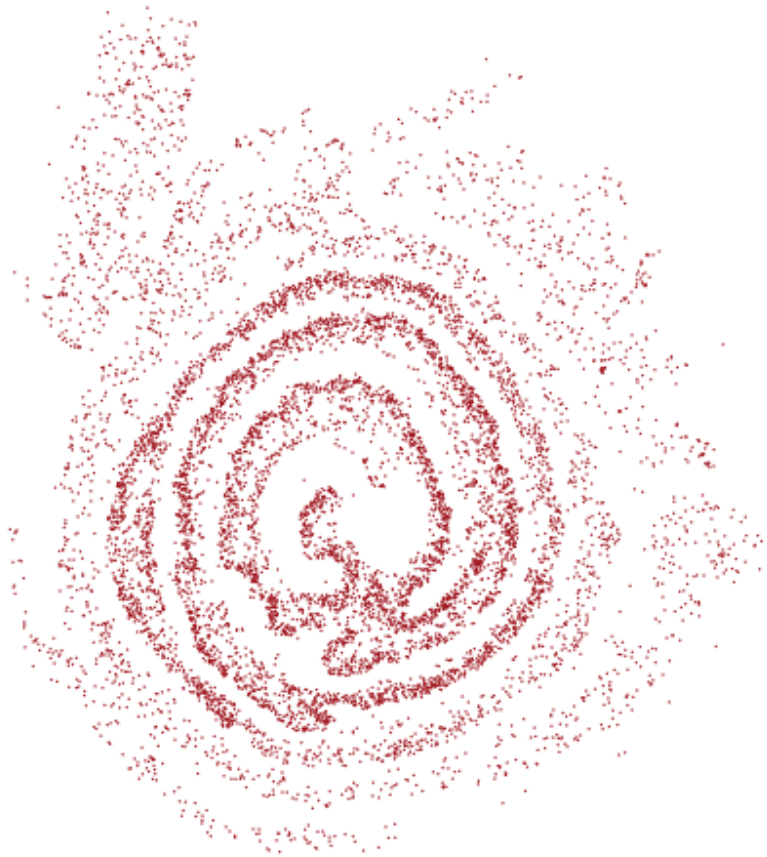
5. Metabolite calling from celltypes

Colonocytes Tryp-

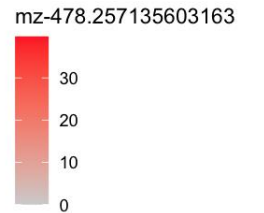
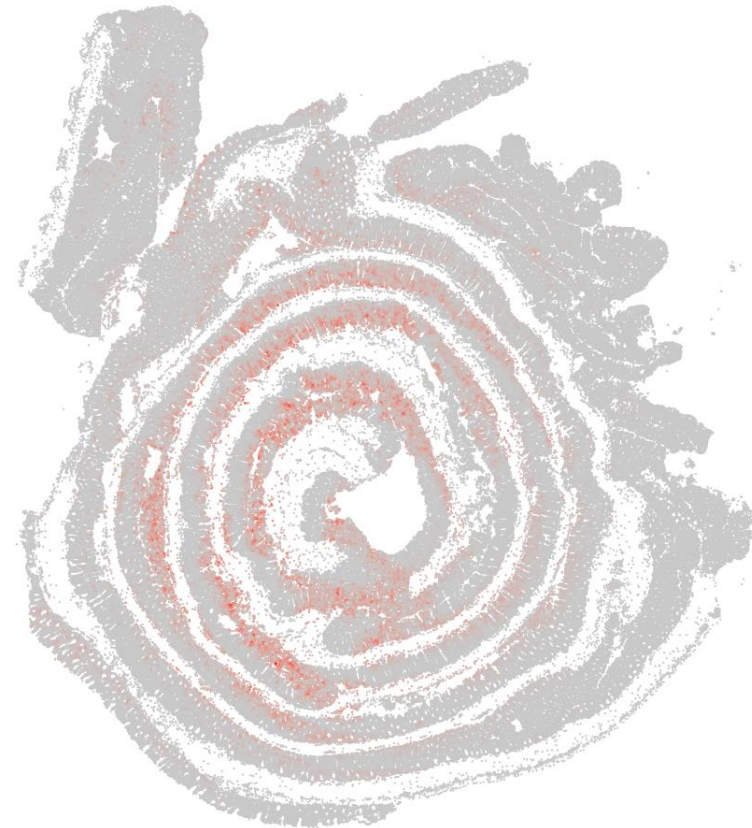


5. Metabolite calling from celltypes

Colonocytes Tryp-



Spatial expression: mz-478.257135603163

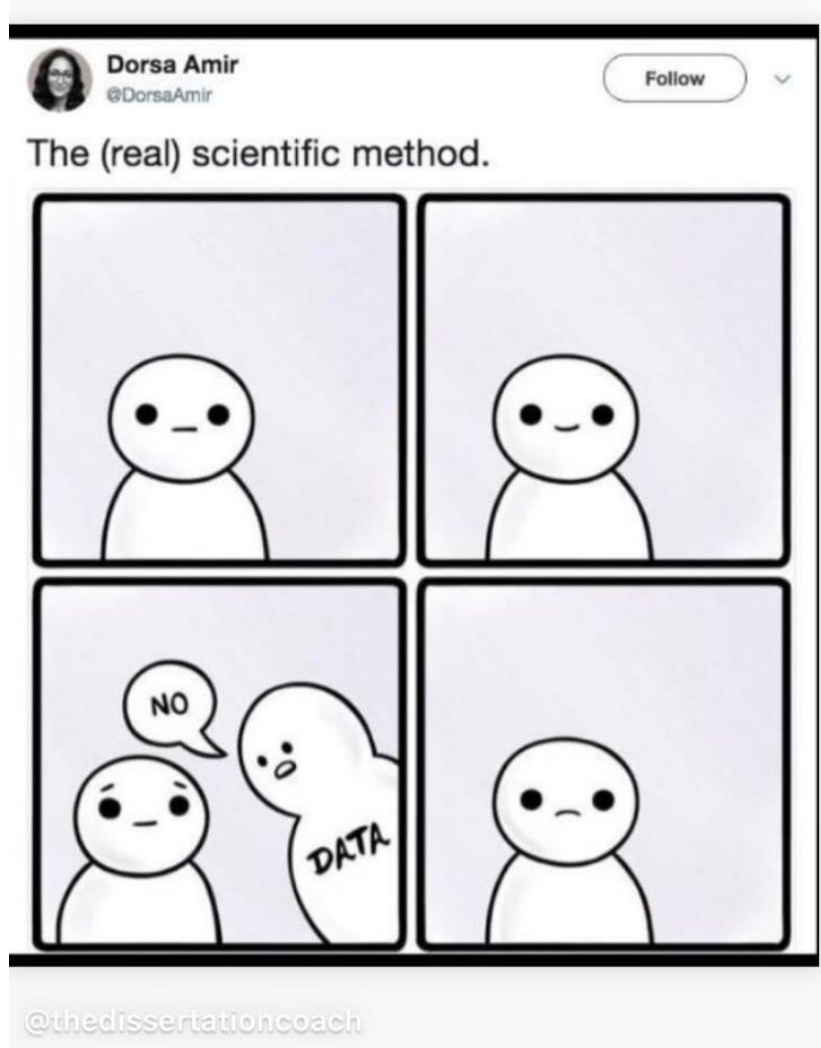


Other steps?

- Gene-metabolite correlation
- Pathway enrichment analysis
- Individual cell-type quantification

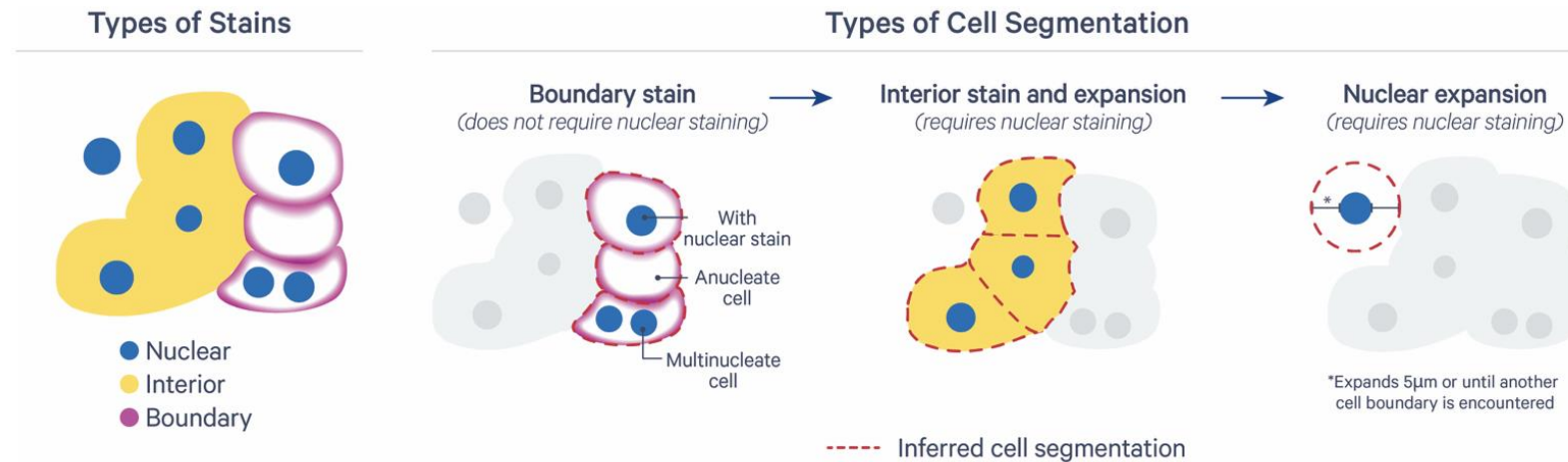
... integration with other layers?

Questions?



Cellular segmentation (10x genomics)

Proprietary algorithm of 10x when using Xenium



Markers: ATP1A1, E-Cadherin, CD45, 18S rRNA marker, DAPI stain and Vimentin

Freely available algorithms: Cellpose (p), StarDist (p), FastReseg (R)